

Frailty Policy Simulation: Method Details

Contents

- I. Verification and Validation (V&V)
- II. Longitudinal results by factor setting
- III. Design of Experiments analysis

I. Verification & Validation (V&V)

1. Table 3 from the paper is reproduced below for a summary.

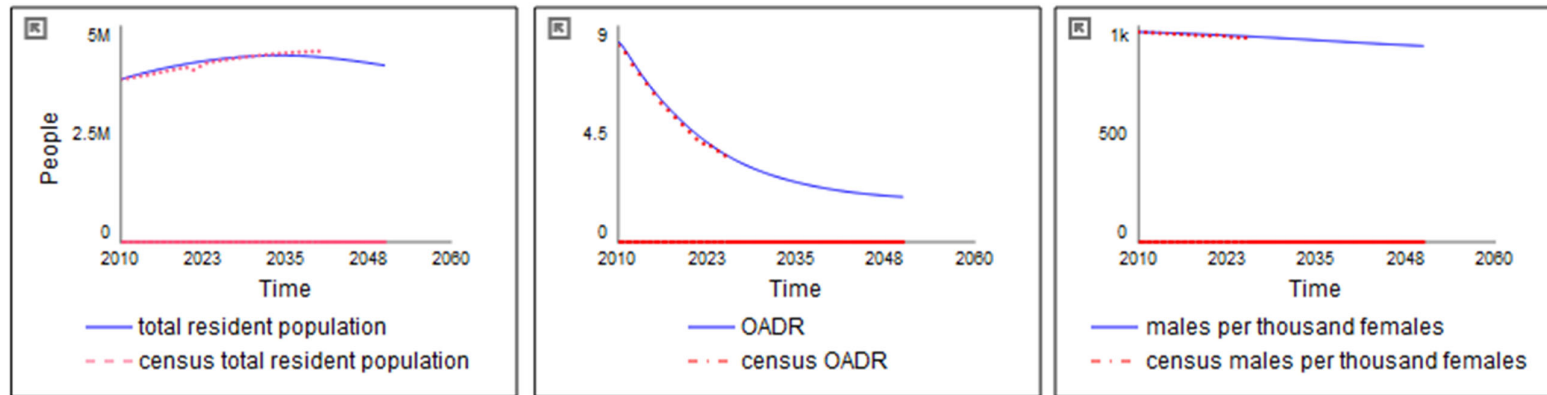
Table 3: V&V steps completed and planned.

	Meaning	Steps taken and planned
Conceptual validation	Model representation of the system is reasonable for the intended purpose	• Study team includes clinicians (BHR, LES) working in the frailty domain in Singapore. • Model logic, inputs, and results were presented to a group of practitioners dedicated to frailty management in June 2026. More reviews will be sought.
Operation validation	Output behavior has sufficient accuracy when compared to real-world	• Population outputs are validated against Singapore census data as well as available Department of Statistics projections for 2026-2040. • Frailty prevalence projections will be validated through further reviews with practitioners.
Specification verification	Software design and the specification for programming and implementing is satisfactory	• Model design has been reviewed by experts within the study team (RZ, SSWL) as well as an external expert.
Implementation verification	Simulation model is implemented according to the simulation model specification	• Model implementation has been reviewed within the team as well as an external expert. • The model is public and the study team welcomes review and improvement suggestions.

2. Population

- Fertility and mortality rates were based on 2010-2025 averages as per census data.
- Stella's model calibration feature was used to estimate immigration rate such that model total resident population aligned with census data for 2010-2025 as well as with the average of the lower and upper bound resident population projection for 2026-2040.

Screenshots from the model showing model versus official figures are shown below.



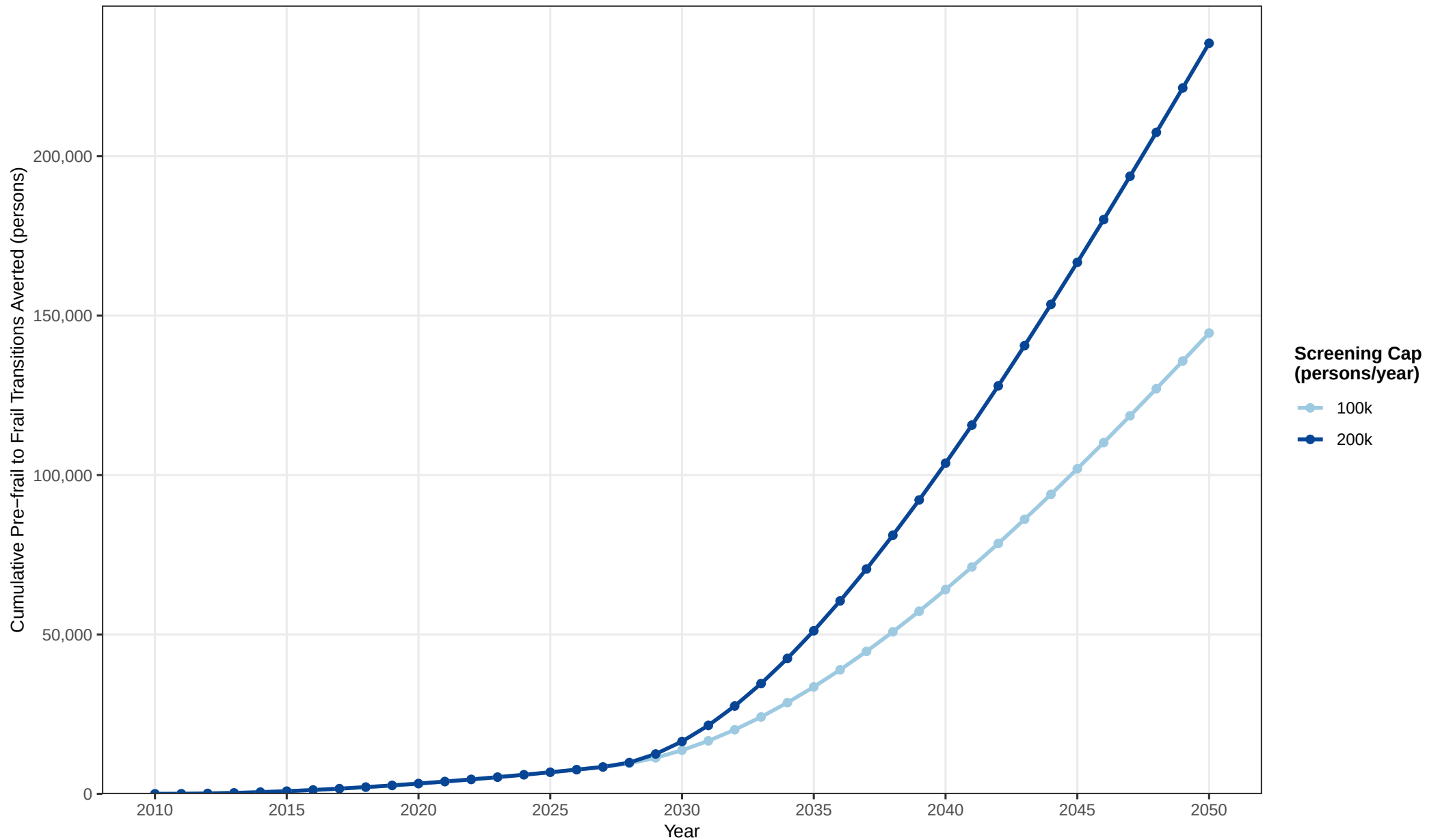
3. Frailty

- Frailty prevalence and incidence rates are based on Singapore studies.
- Some parameters (relative risk of mortality and of progression from pre-frail to frail for patients who are well-managed) were taken from global studies. Low and high values of these parameters (based on inter-quartile range) were used as factor settings in the analysis.
- Low and high values of 50% and 70% for the input variable for proportion of patients well-managed were based on discussion with a group of frailty practitioners.
- For further validation, the study team will continue to engage with stakeholders and seek current frailty prevalence statistics.

II. Longitudinal results by factor setting

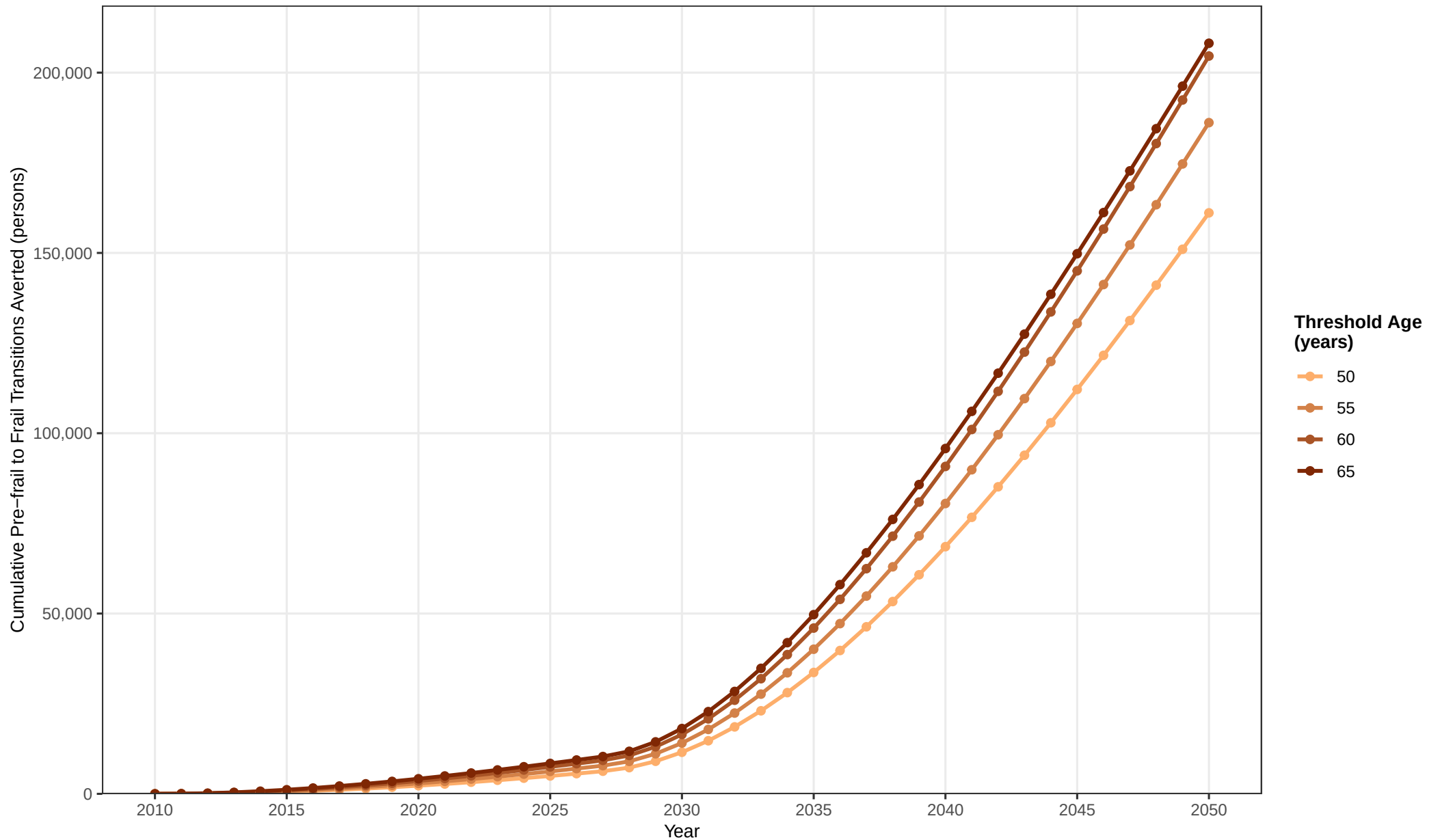
Cumulative Pre-frail to Frail Transitions Averted (persons)

Mean by Screening Cap (Universal) setting



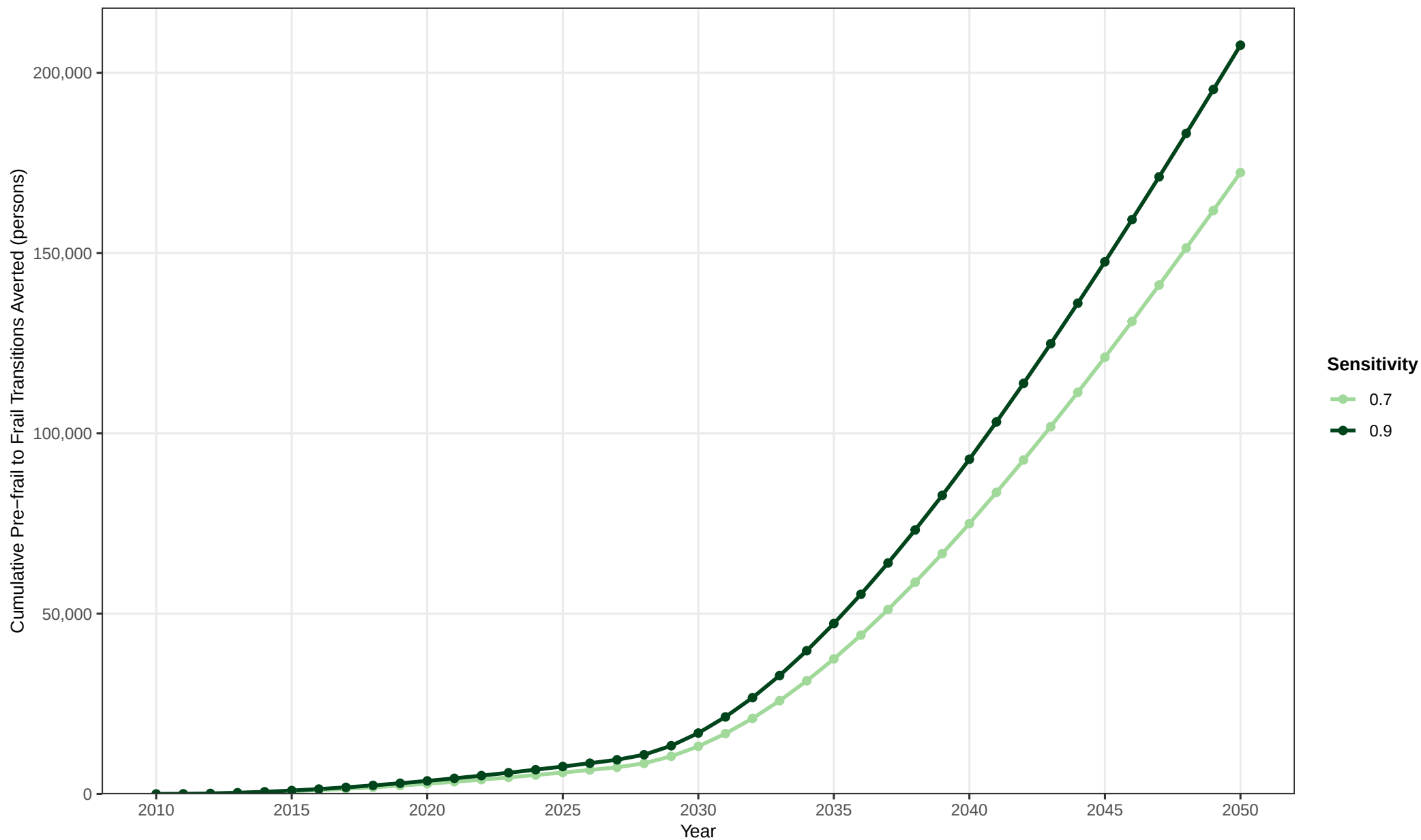
Cumulative Pre-frail to Frail Transitions Averted (persons)

Mean by Screening Threshold Age setting



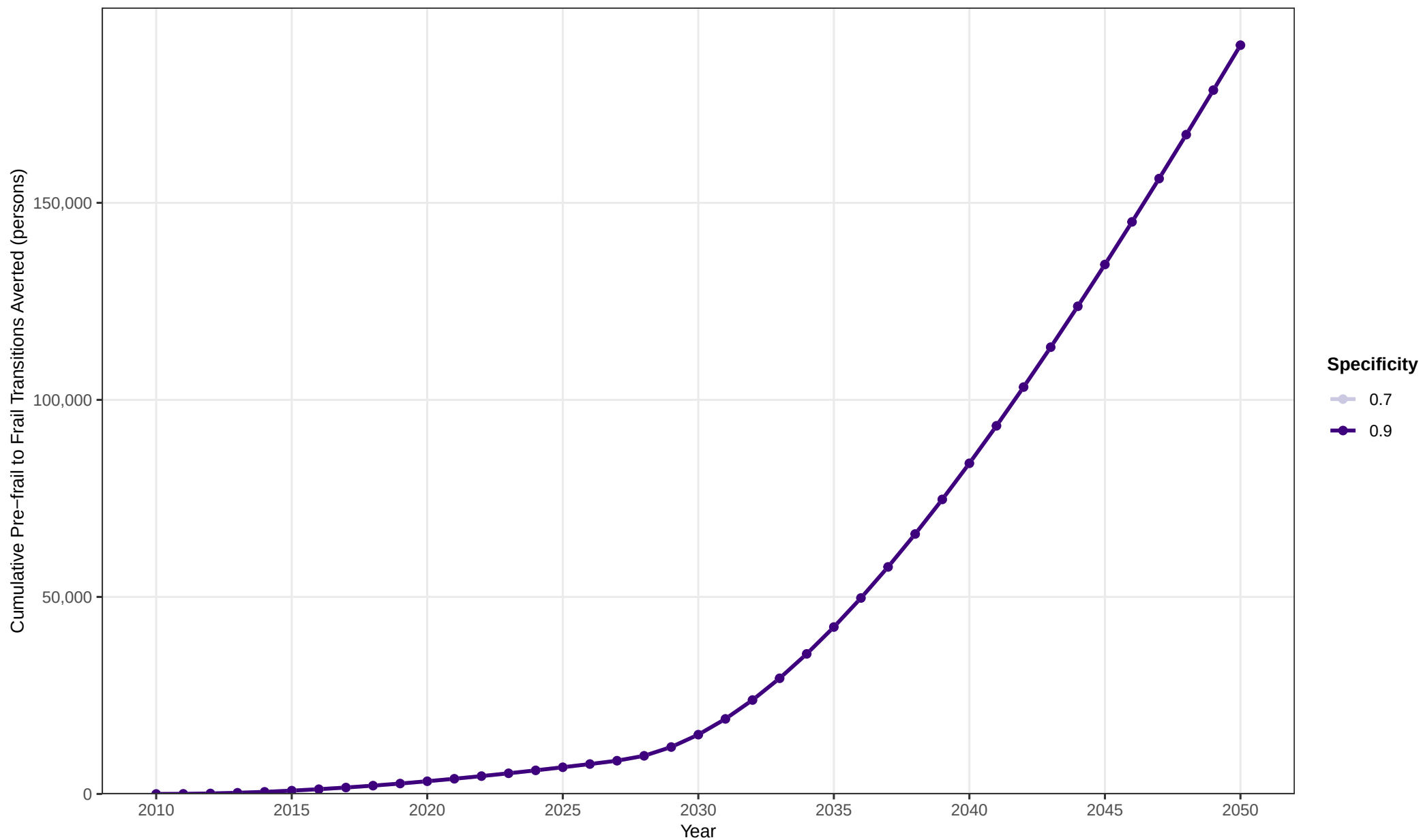
Cumulative Pre-frail to Frail Transitions Averted (persons)

Mean by Screening Sensitivity setting



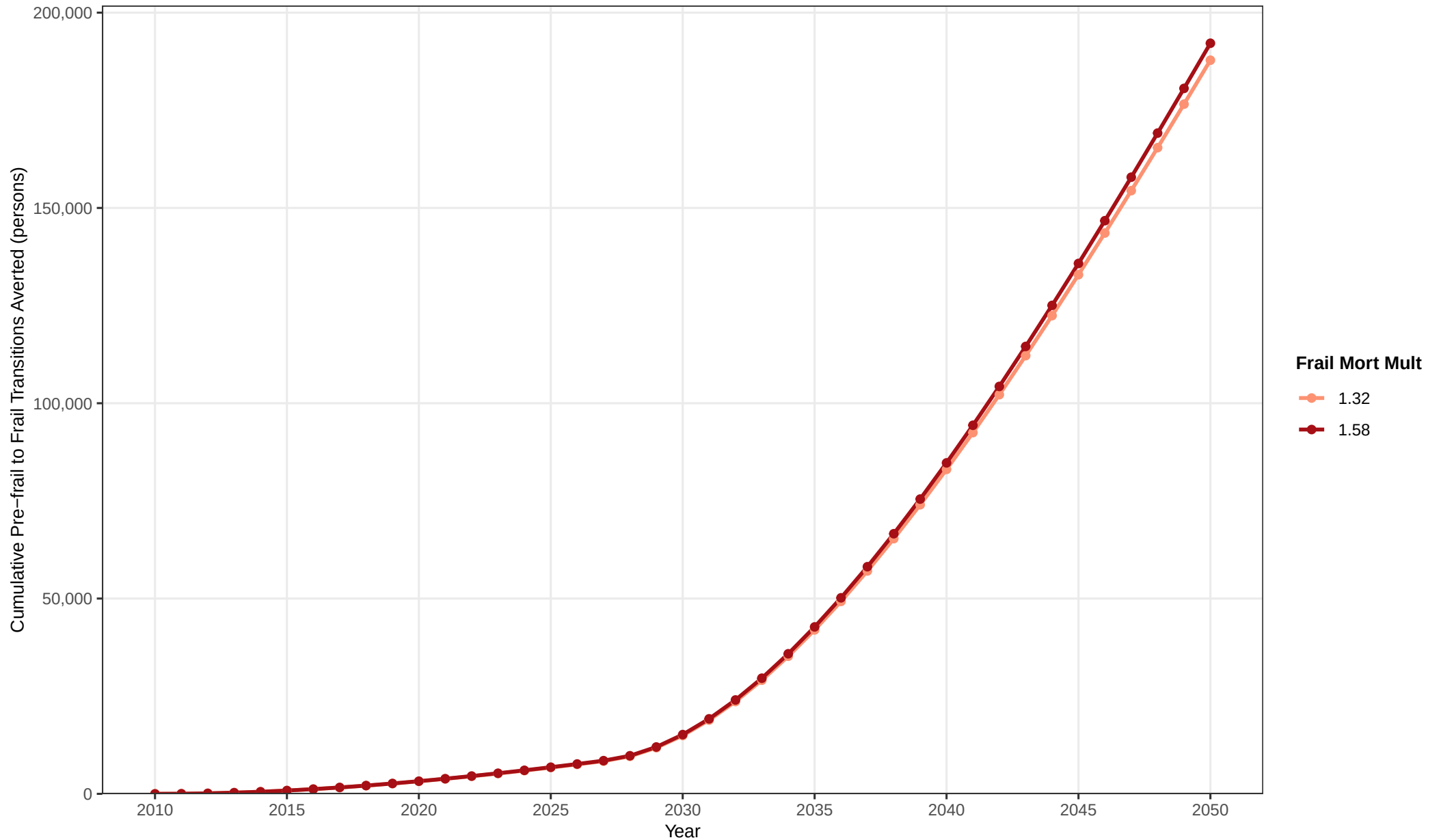
Cumulative Pre-frail to Frail Transitions Averted (persons)

Mean by Screening Specificity setting



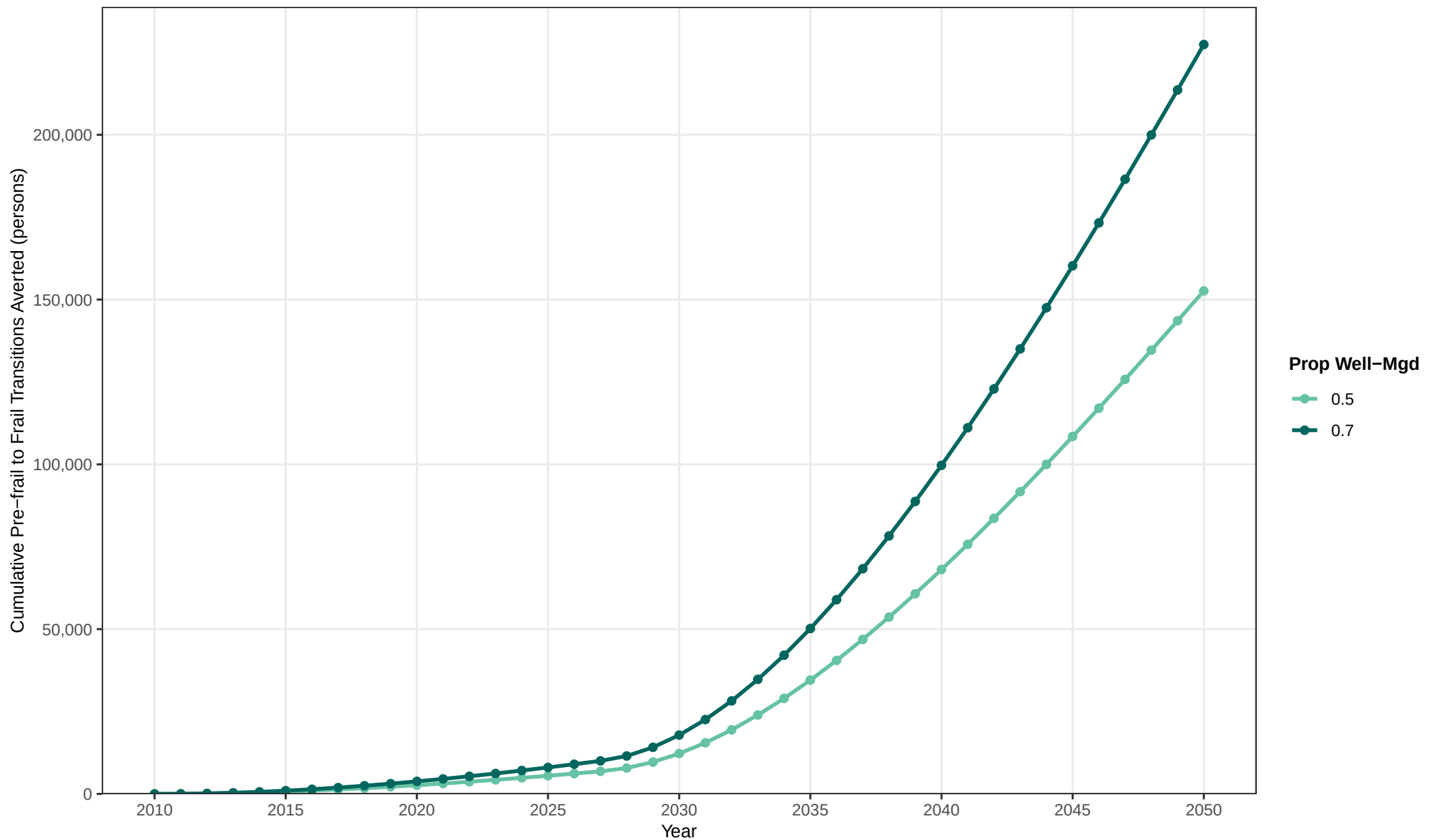
Cumulative Pre-frail to Frail Transitions Averted (persons)

Mean by Frail mortality multiplier setting



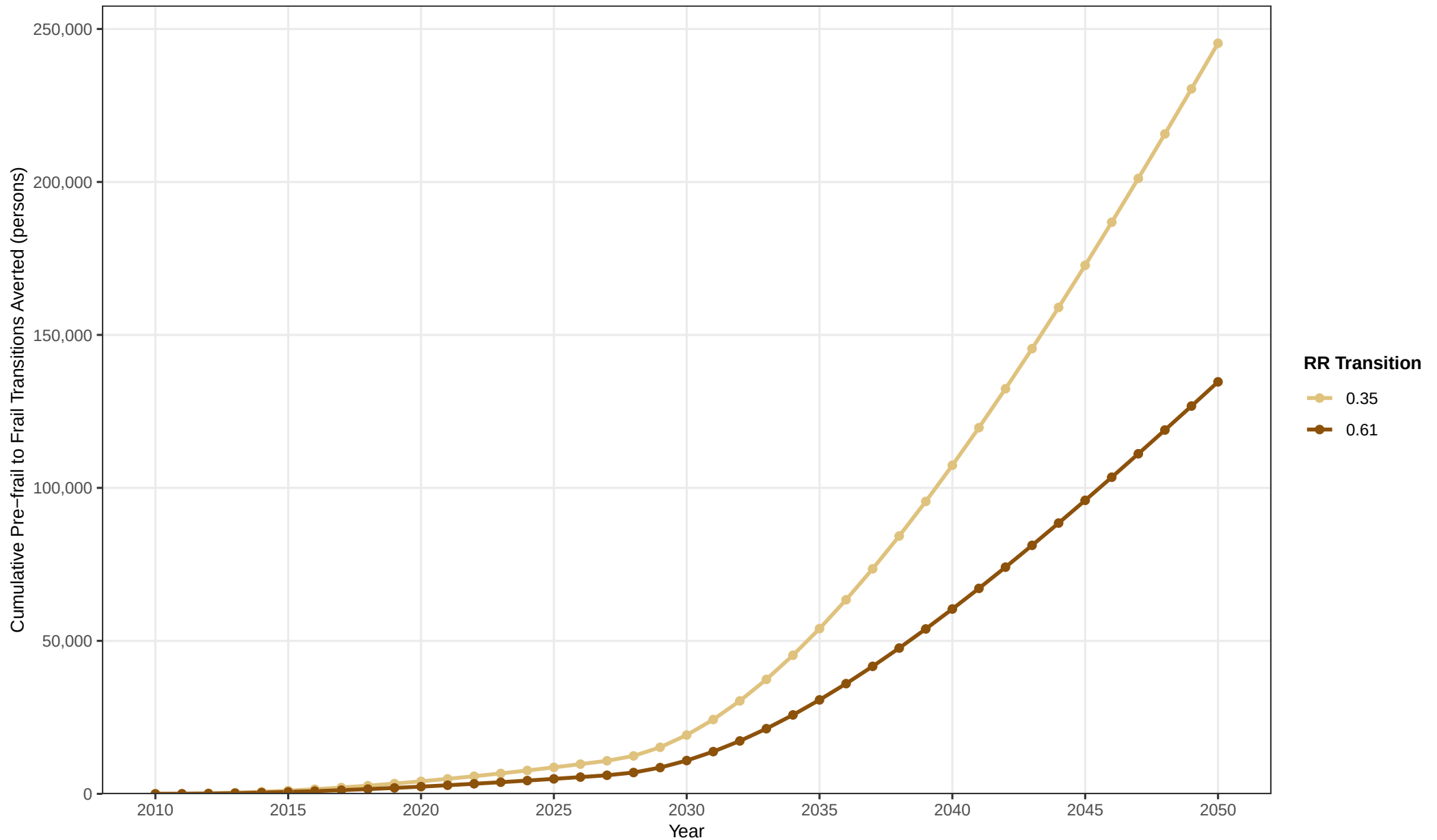
Cumulative Pre-frail to Frail Transitions Averted (persons)

Mean by Proportion well-managed setting



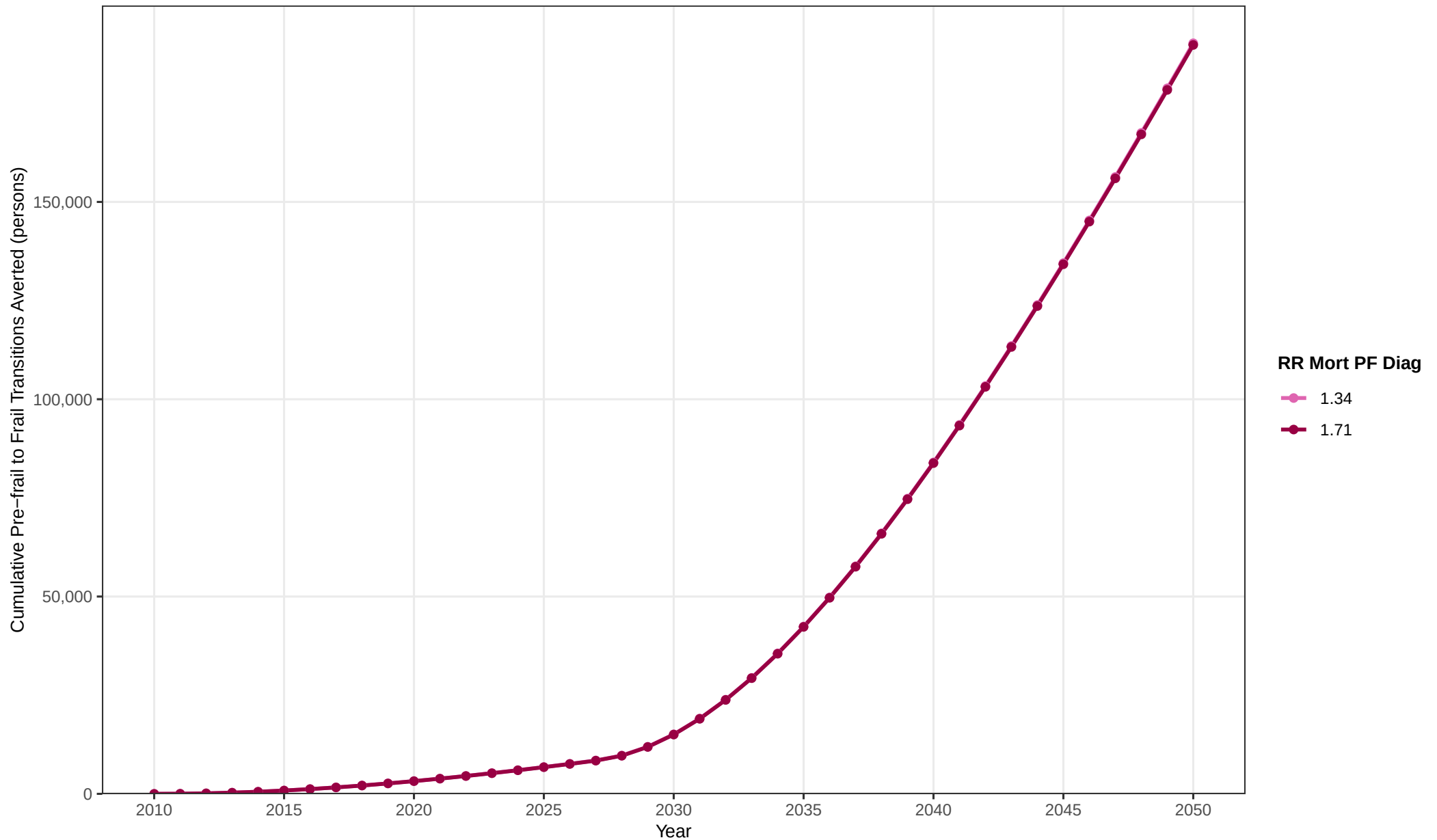
Cumulative Pre-frail to Frail Transitions Averted (persons)

Mean by Relative risk of transition pre-frail to frail setting



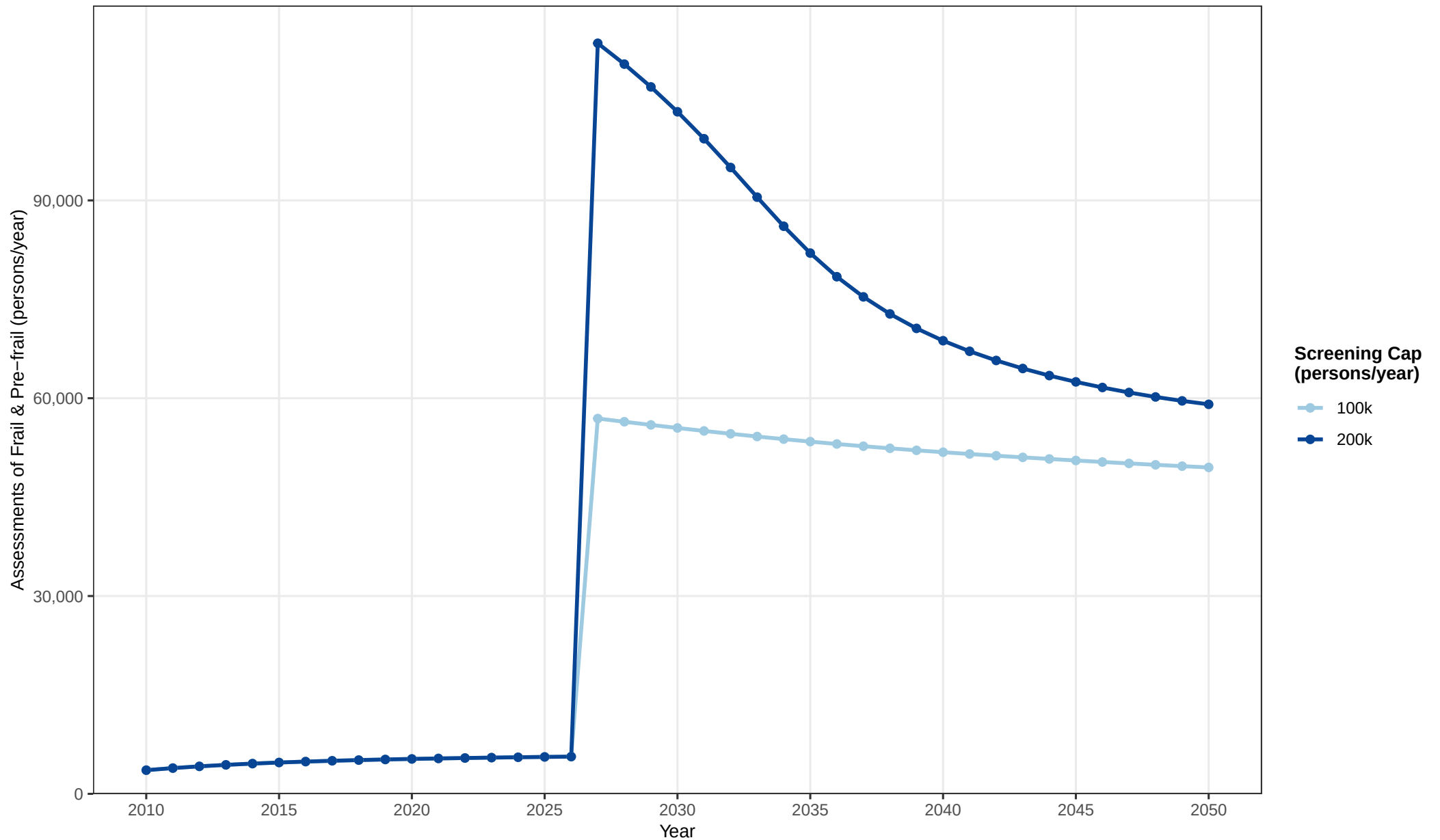
Cumulative Pre-frail to Frail Transitions Averted (persons)

Mean by Relative Risk of mortality for pre-frail setting



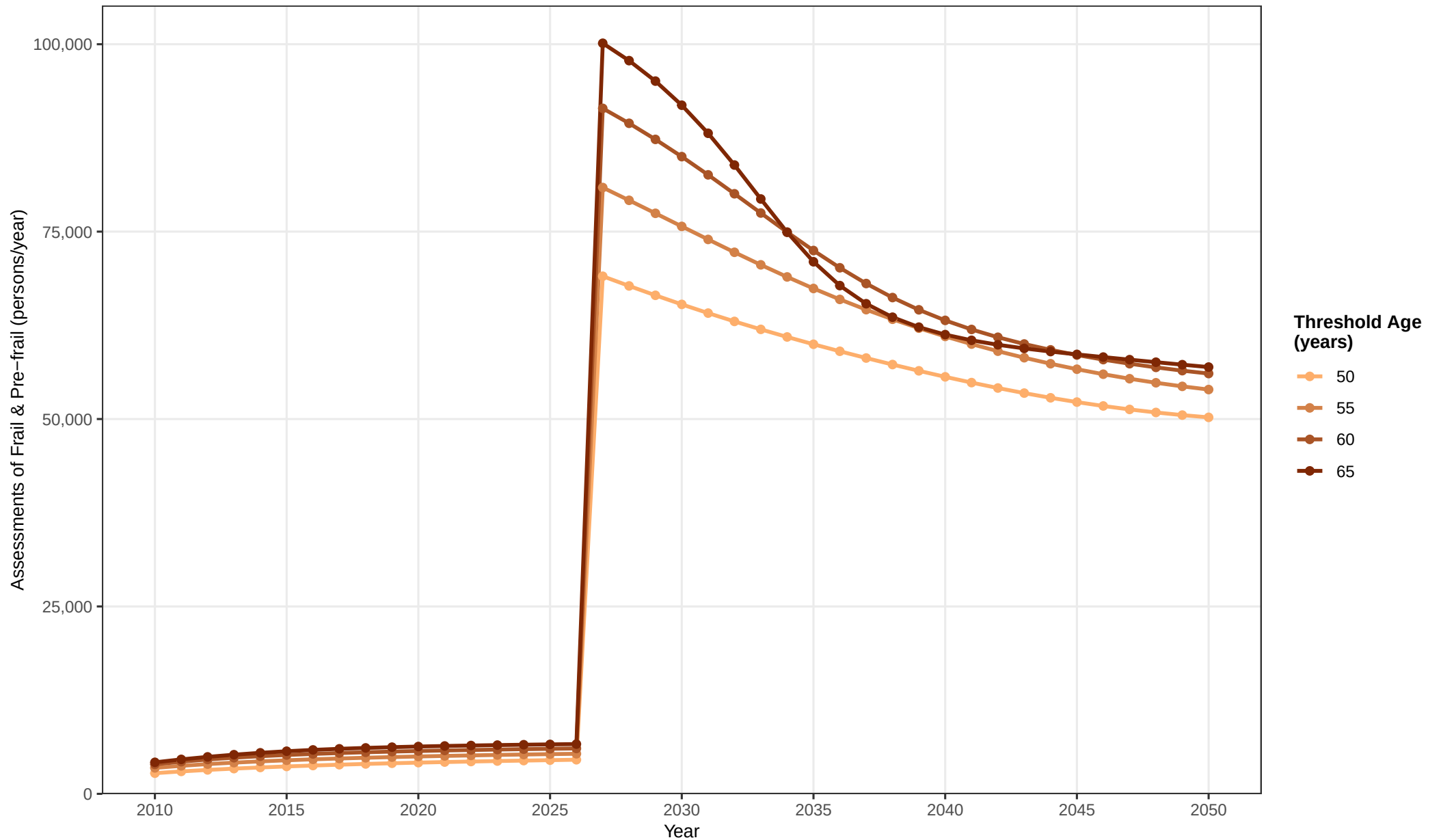
Assessments of Frail & Pre-frail (persons/year)

Mean by Screening Cap (Universal) setting



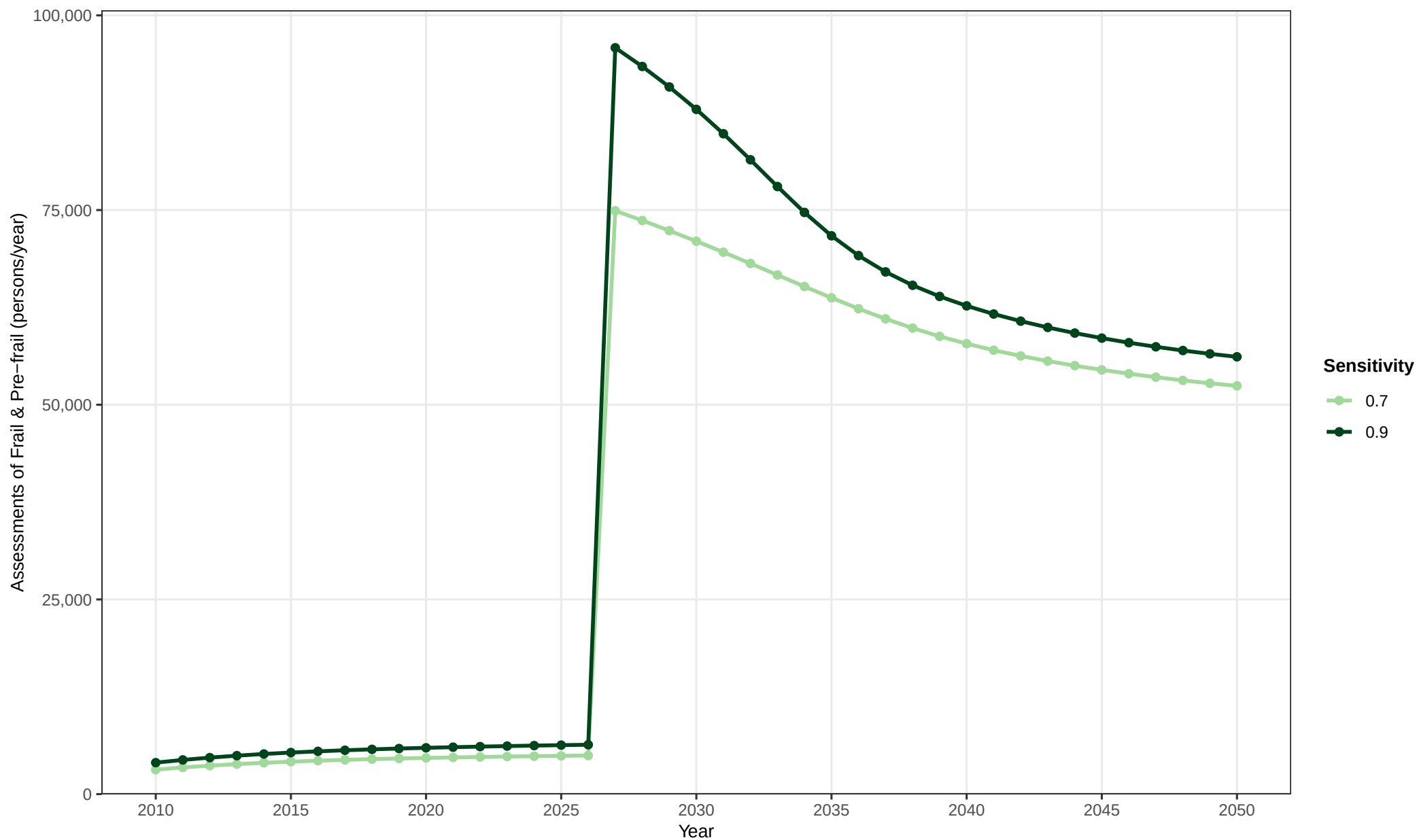
Assessments of Frail & Pre-frail (persons/year)

Mean by Screening Threshold Age setting



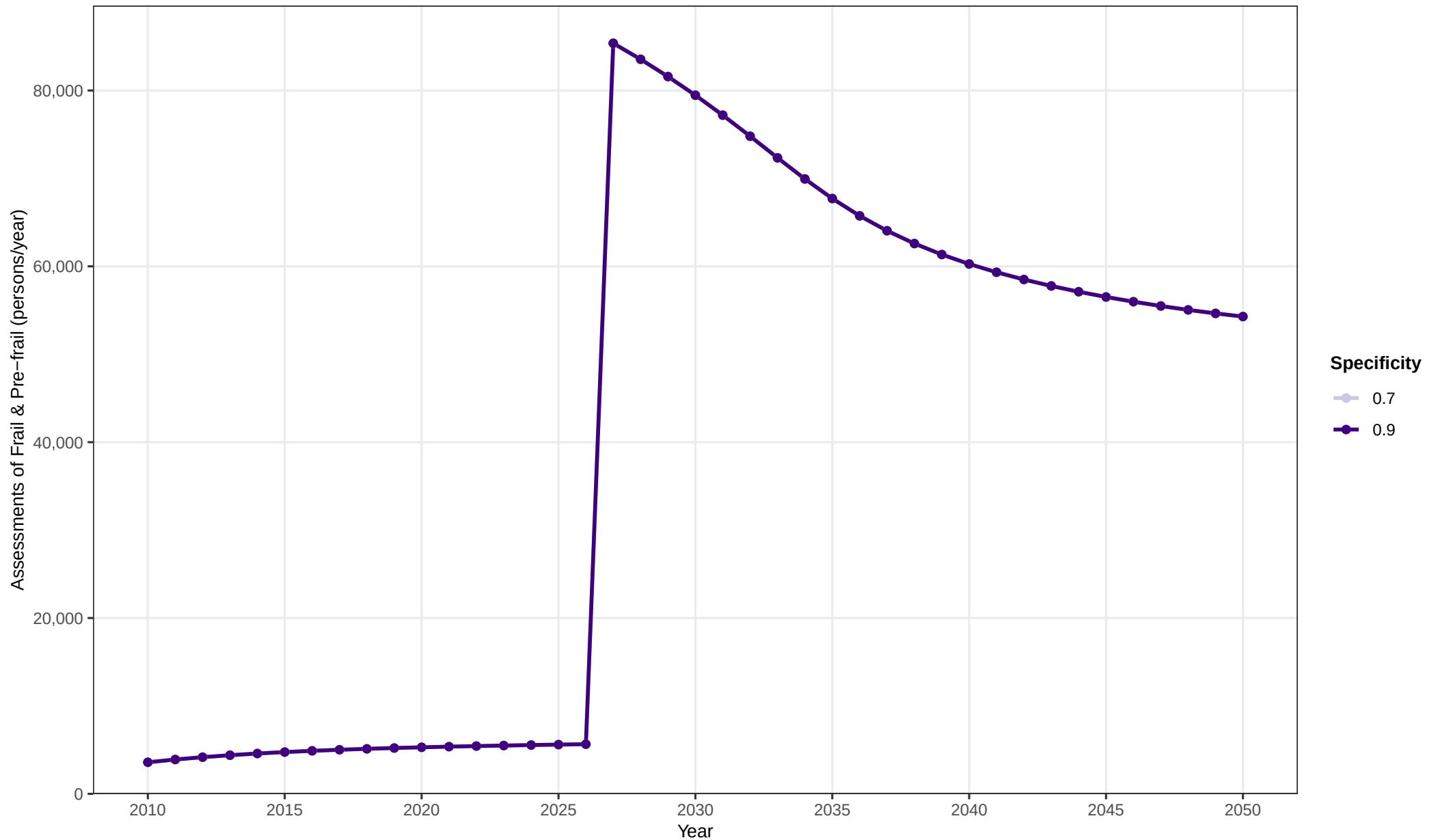
Assessments of Frail & Pre-frail (persons/year)

Mean by Screening Sensitivity setting



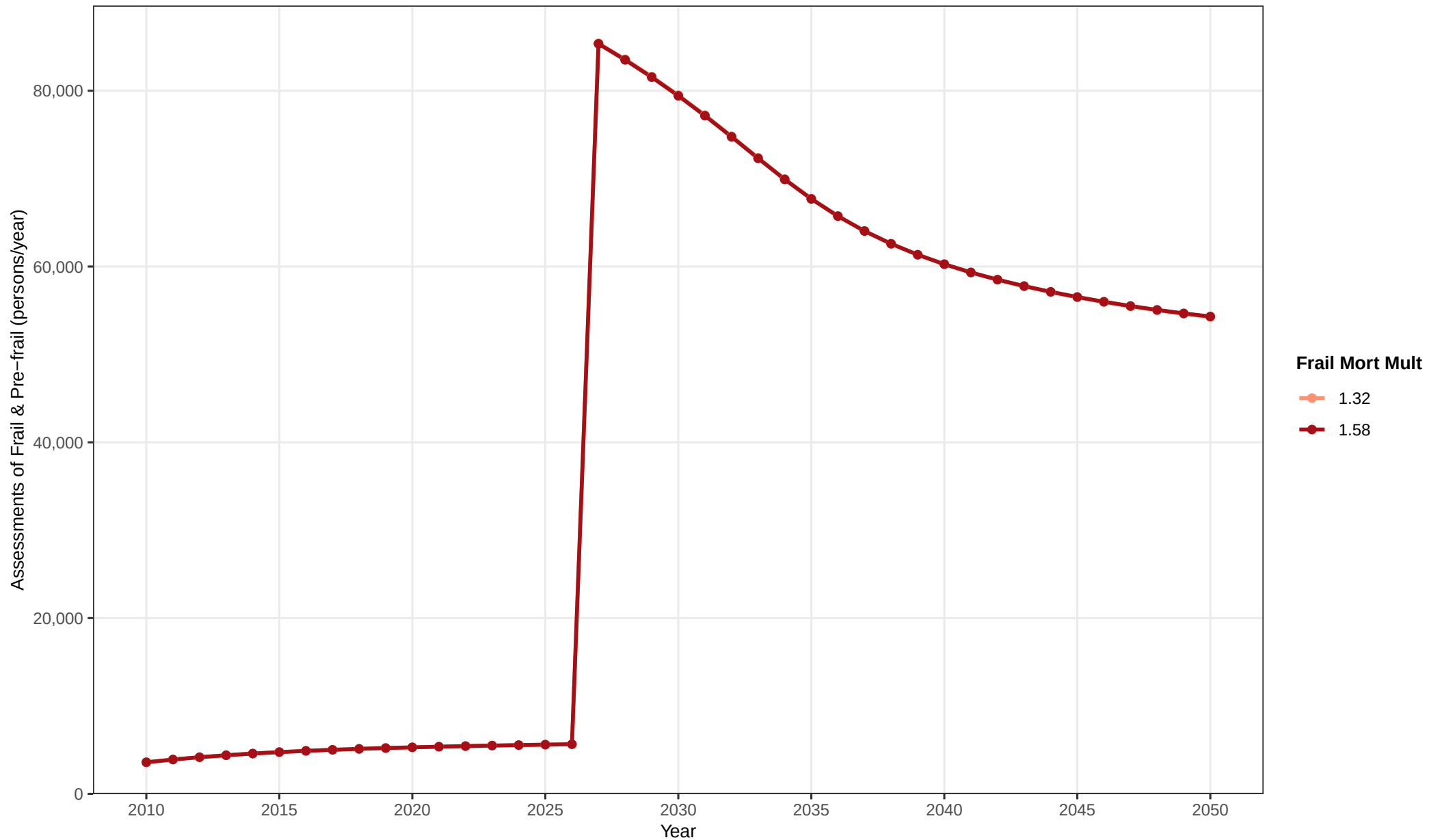
Assessments of Frail & Pre-frail (persons/year)

Mean by Screening Specificity setting



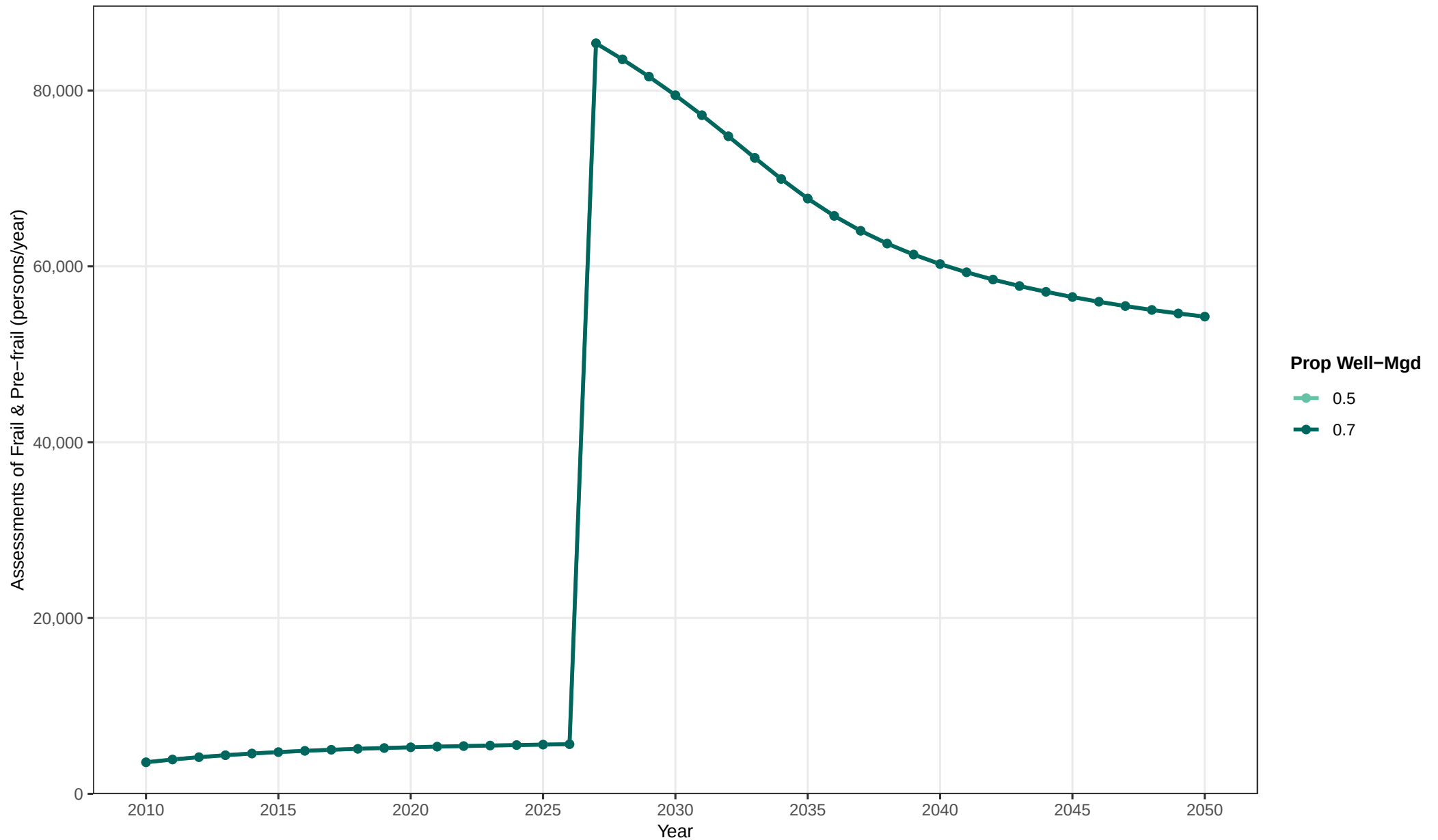
Assessments of Frail & Pre-frail (persons/year)

Mean by Frail mortality multiplier setting



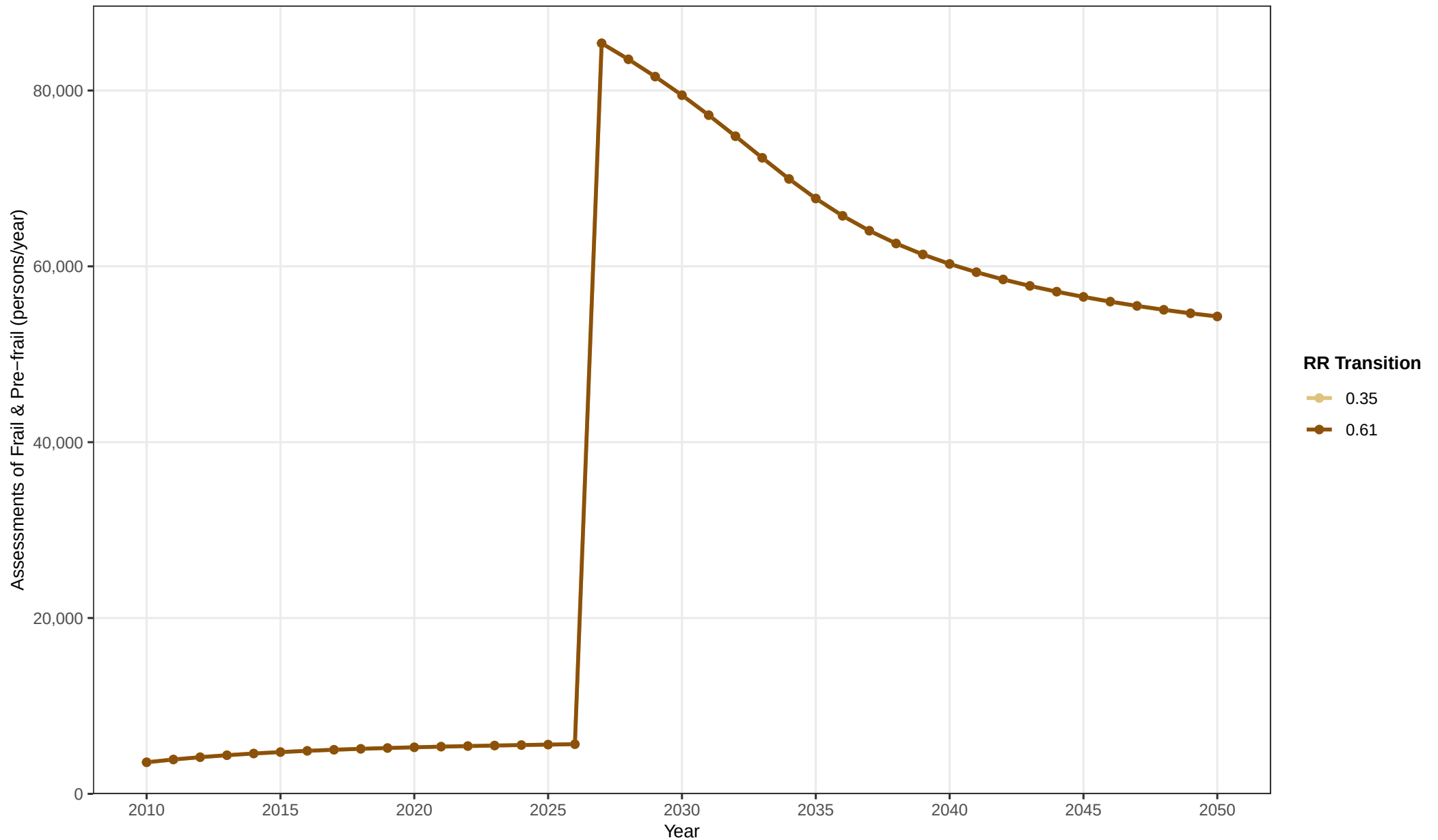
Assessments of Frail & Pre-frail (persons/year)

Mean by Proportion well-managed setting



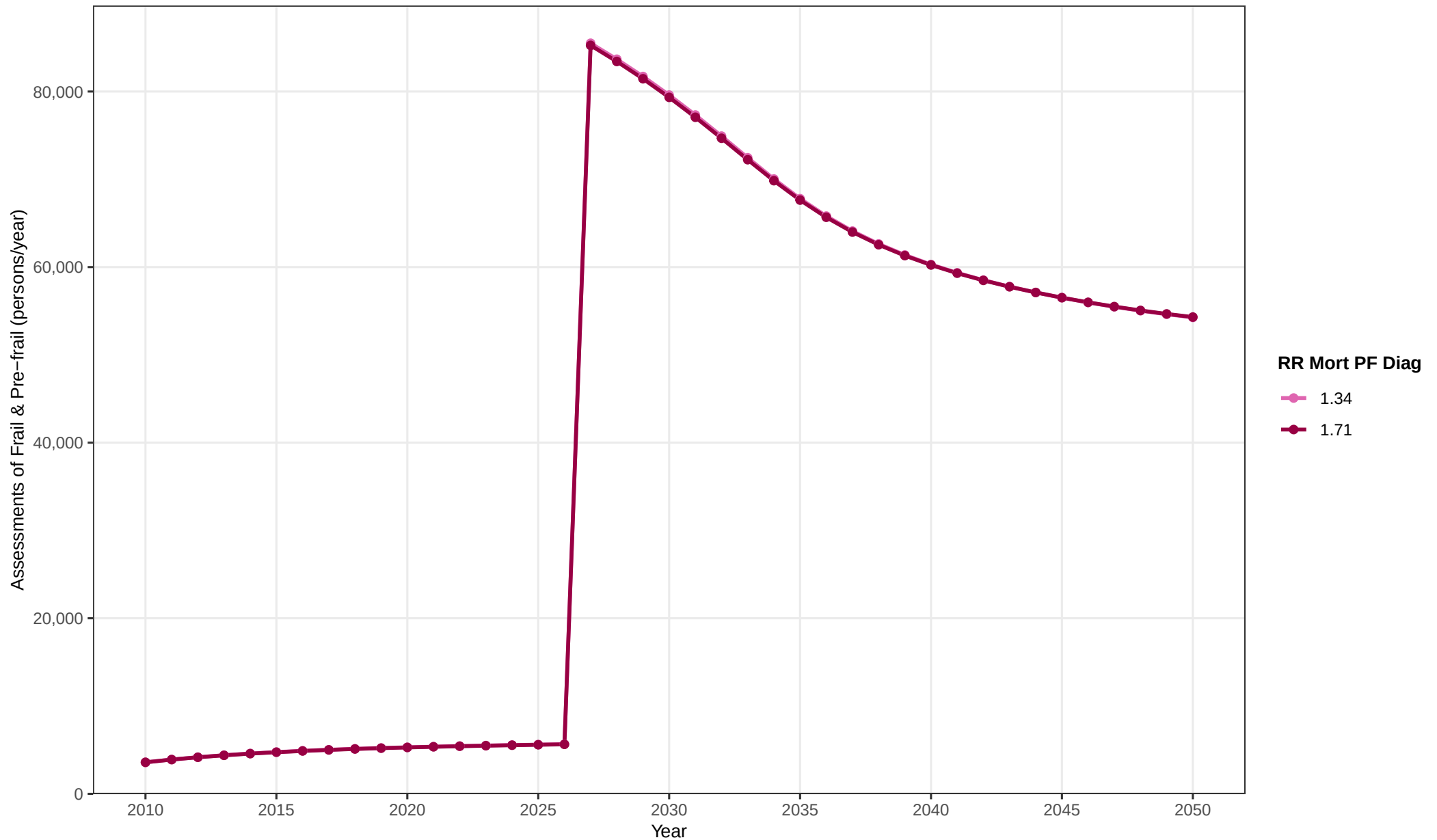
Assessments of Frail & Pre-frail (persons/year)

Mean by Relative risk of transition pre-frail to frail setting



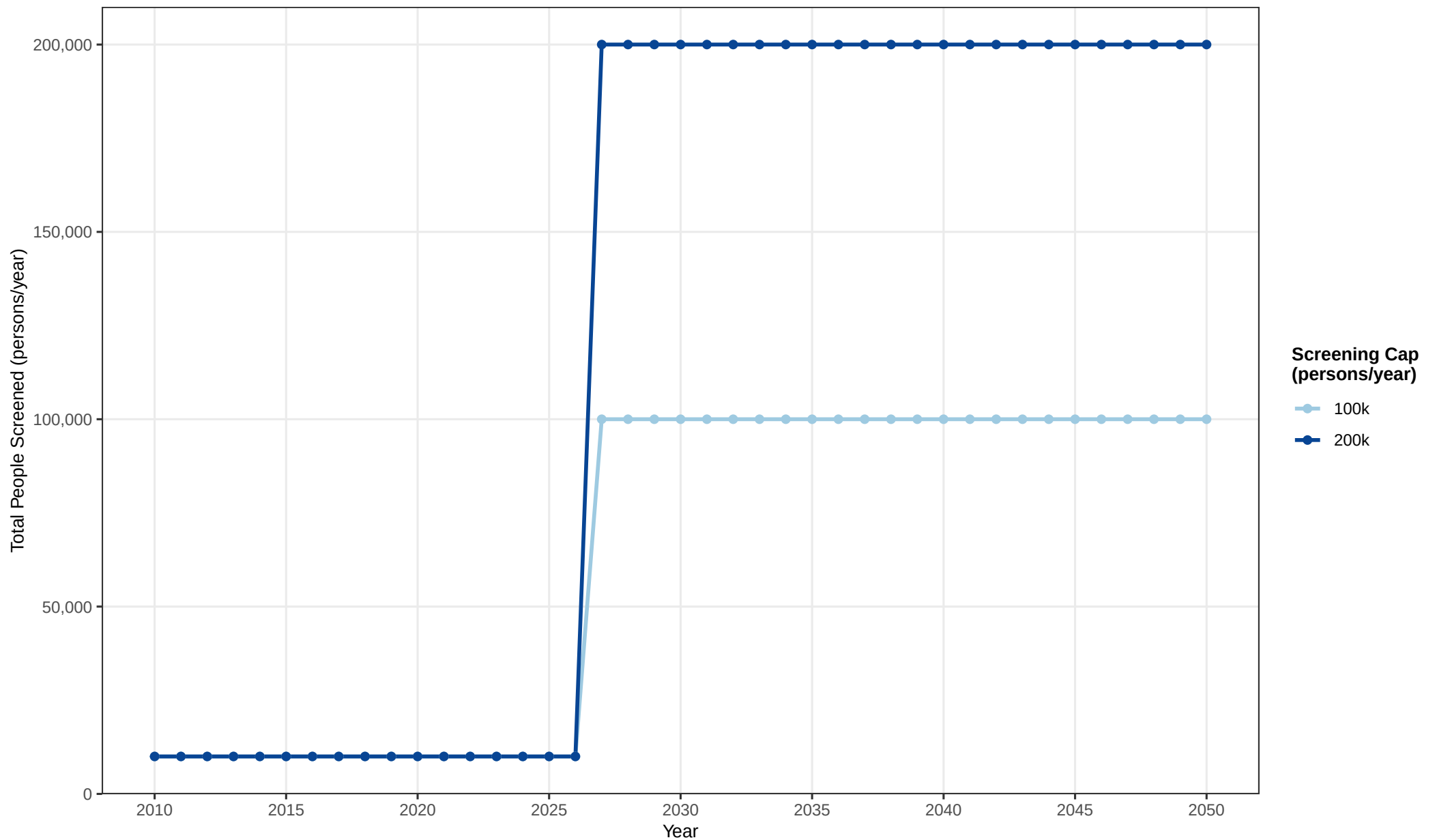
Assessments of Frail & Pre-frail (persons/year)

Mean by Relative Risk of mortality for pre-frail setting



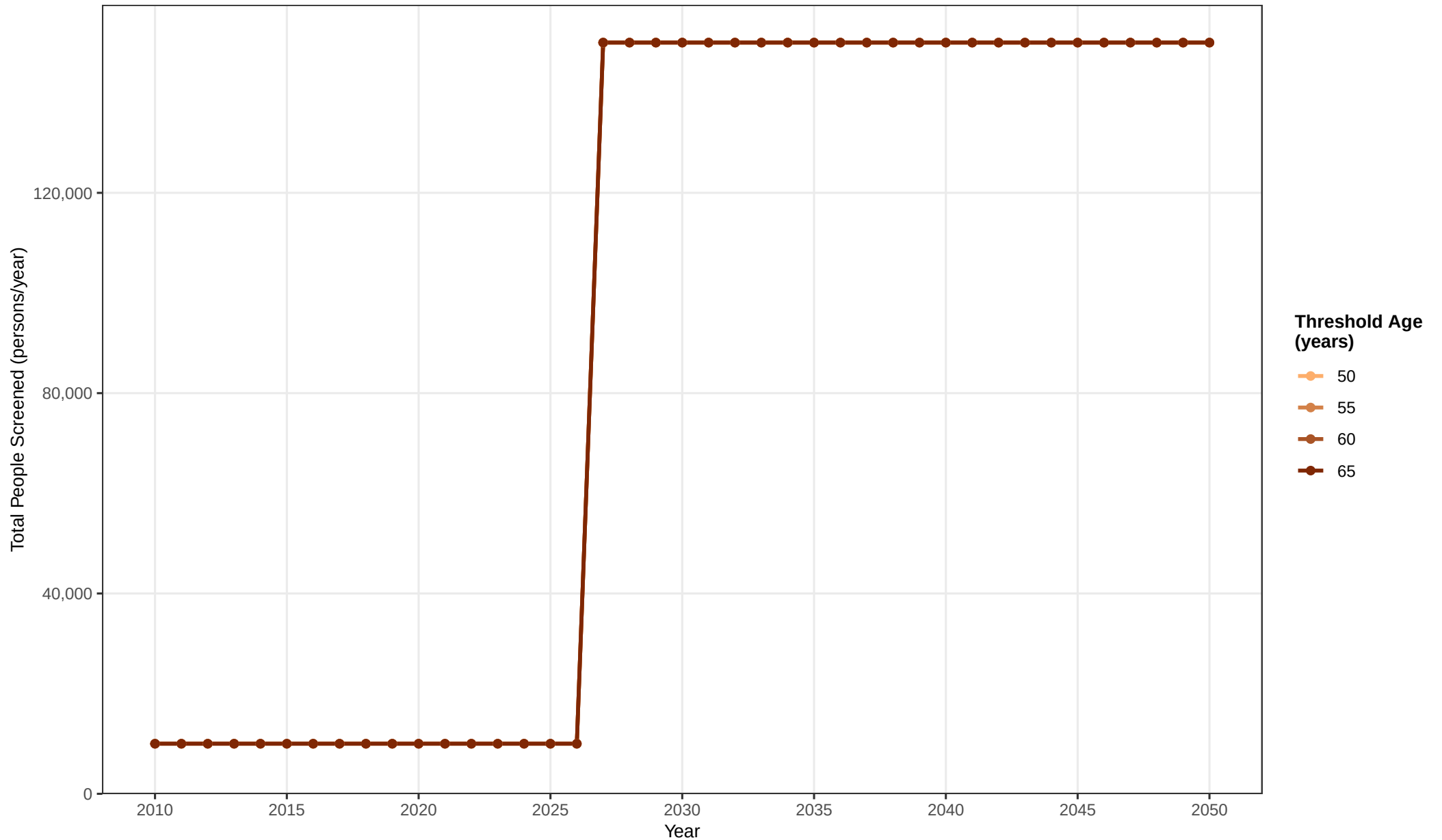
Total People Screened (persons/year)

Mean by Screening Cap (Universal) setting



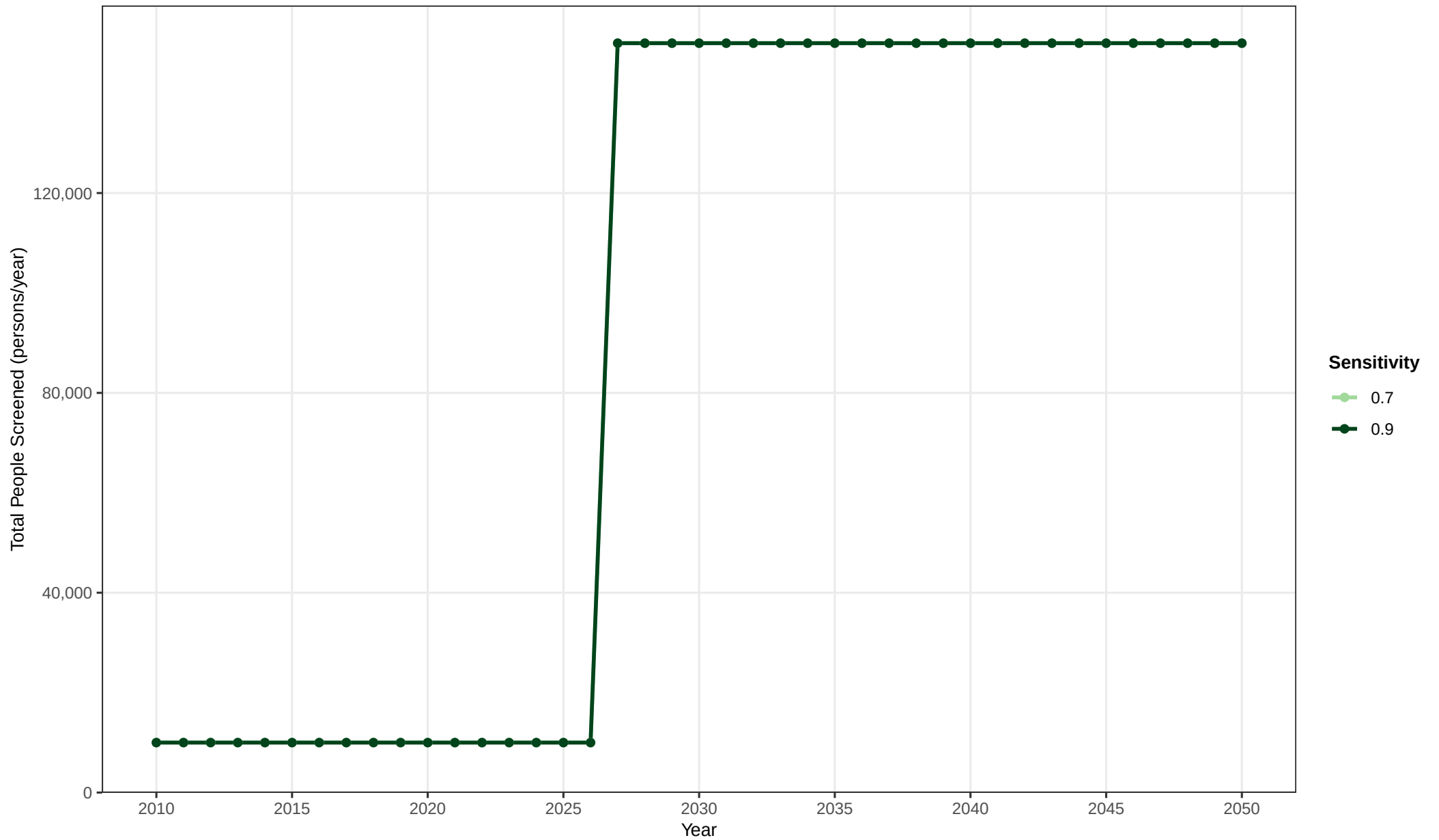
Total People Screened (persons/year)

Mean by Screening Threshold Age setting



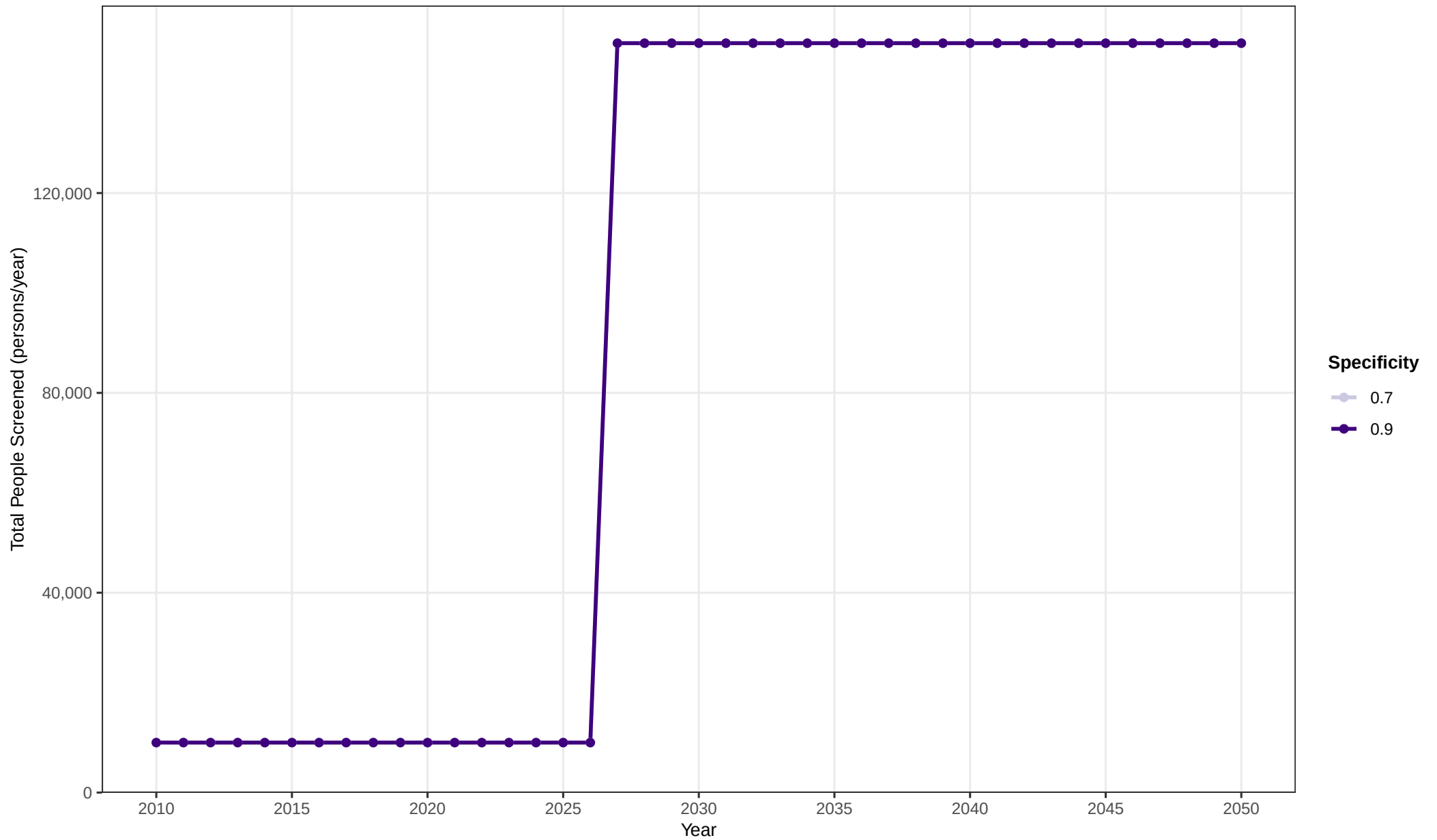
Total People Screened (persons/year)

Mean by Screening Sensitivity setting



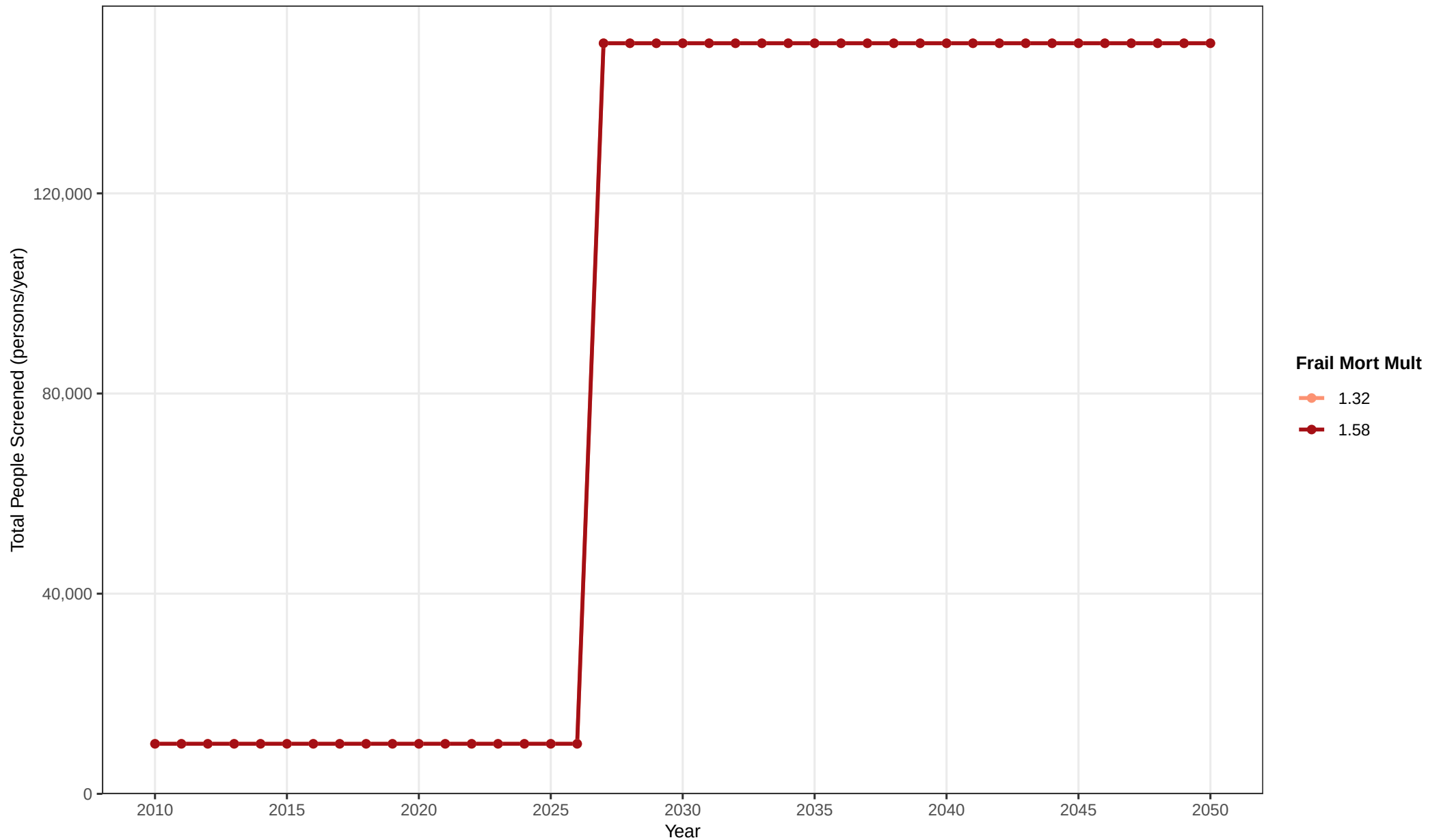
Total People Screened (persons/year)

Mean by Screening Specificity setting



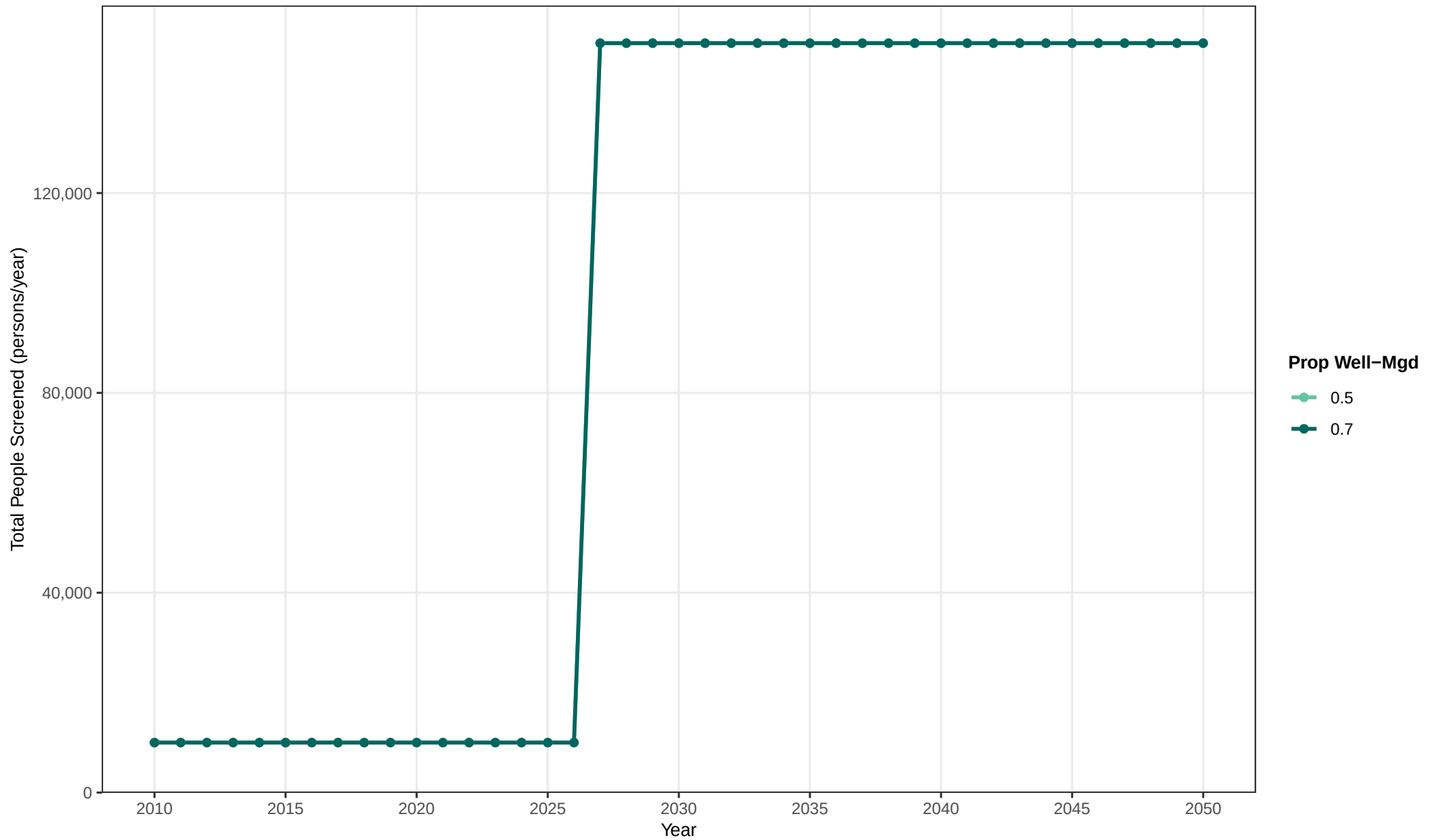
Total People Screened (persons/year)

Mean by Frail mortality multiplier setting



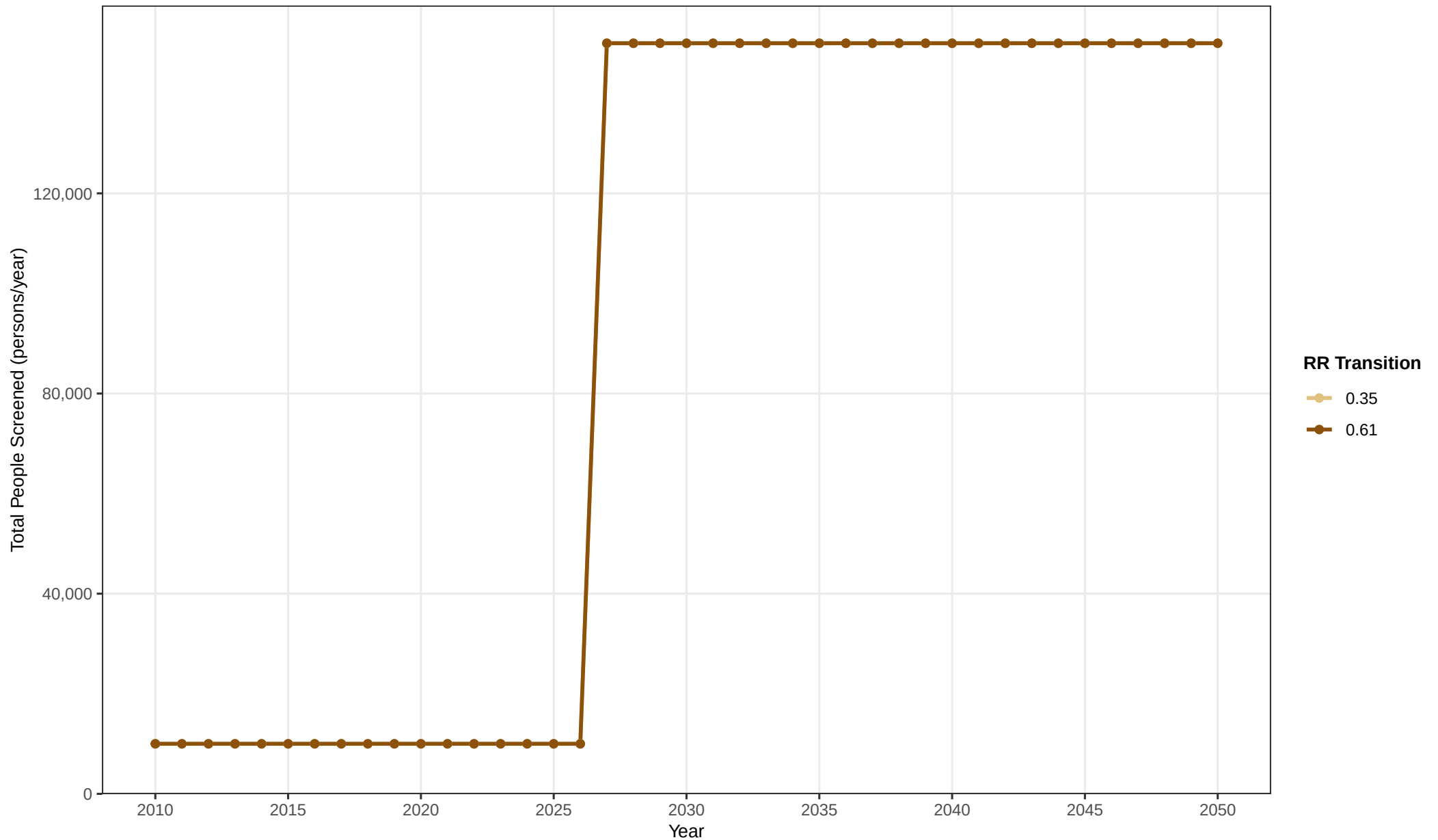
Total People Screened (persons/year)

Mean by Proportion well-managed setting

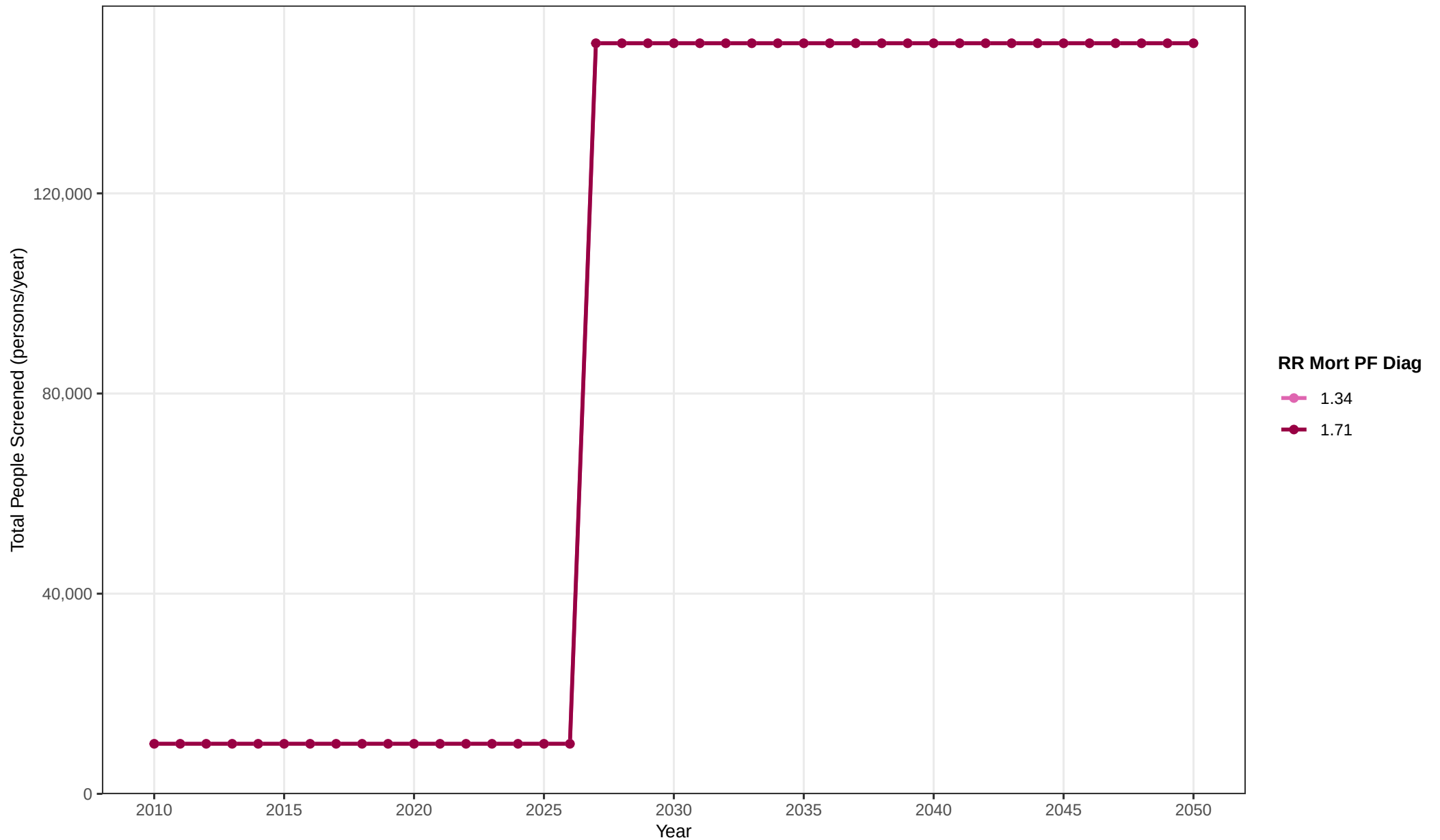


Total People Screened (persons/year)

Mean by Relative risk of transition pre-frail to frail setting

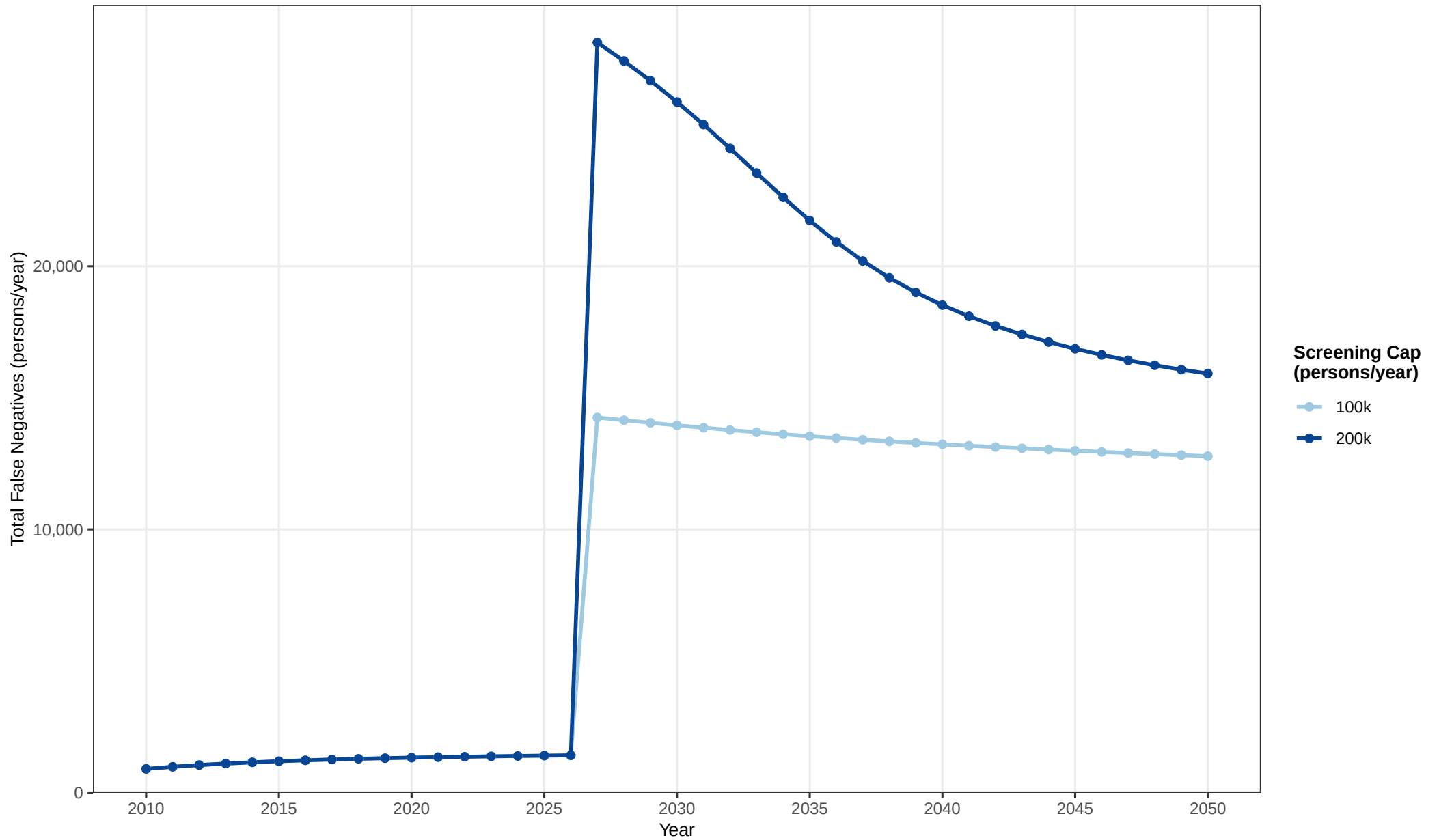


Total People Screened (persons/year)
Mean by Relative Risk of mortality for pre-frail setting



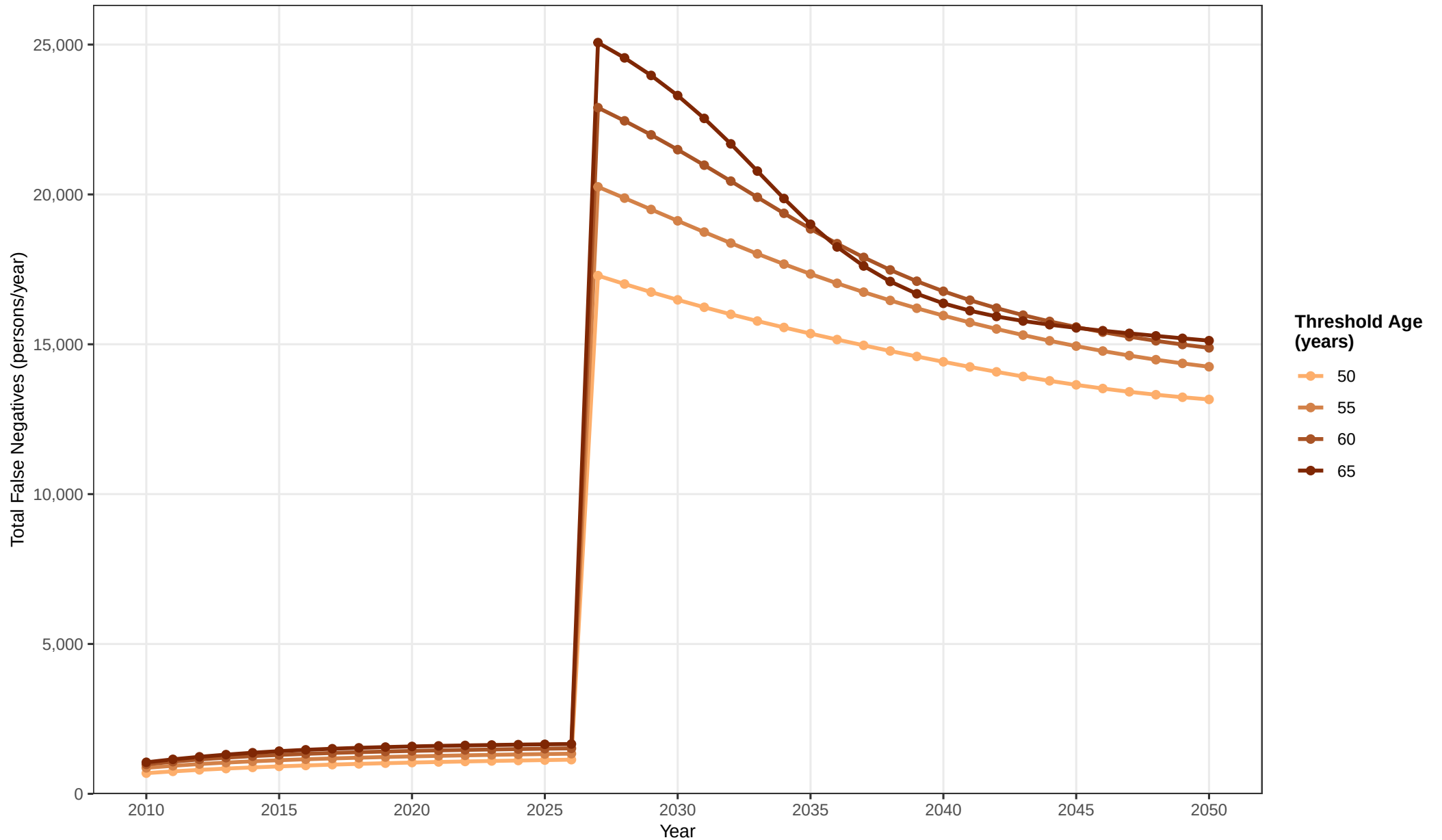
Total False Negatives (persons/year)

Mean by Screening Cap (Universal) setting



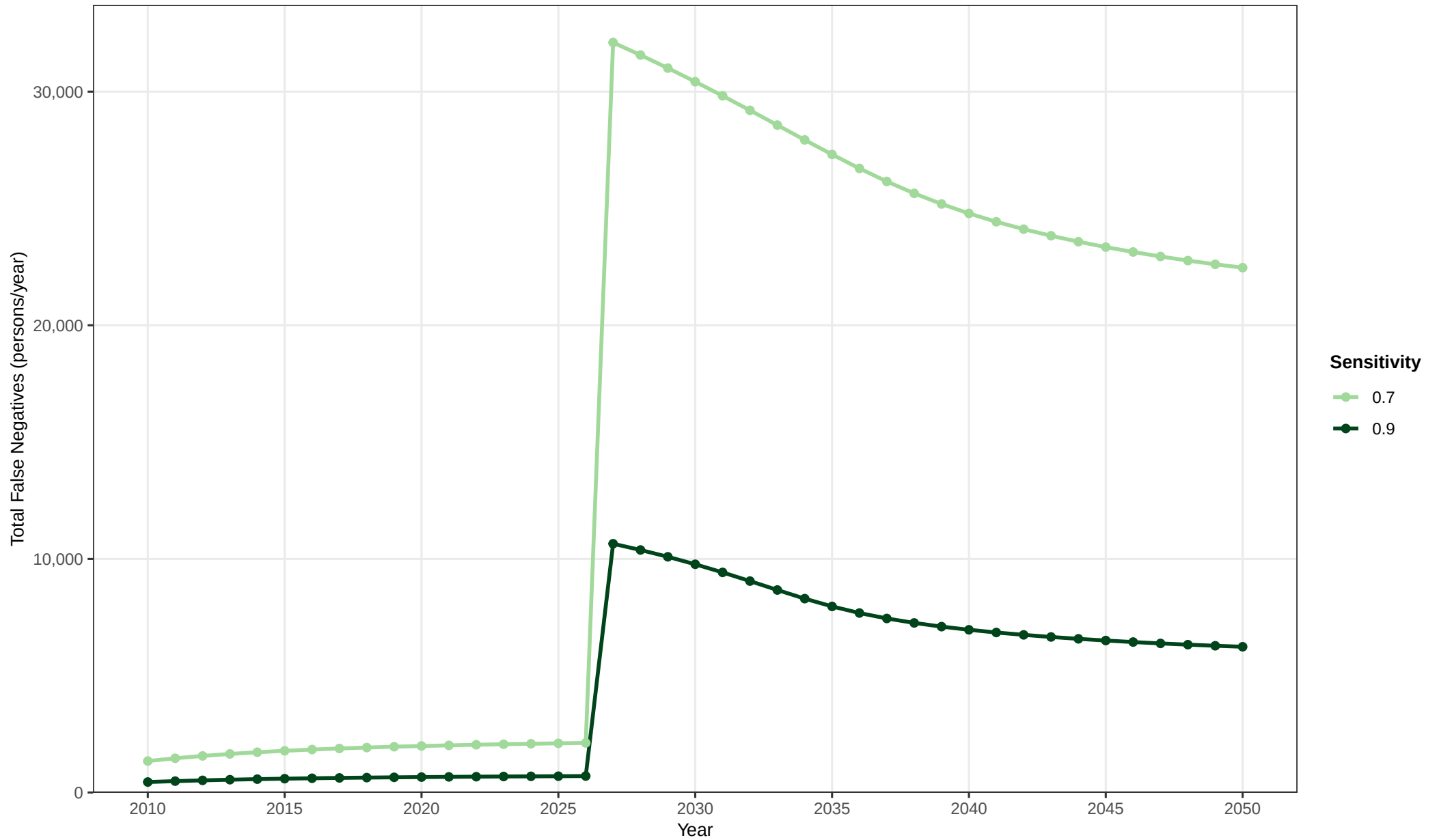
Total False Negatives (persons/year)

Mean by Screening Threshold Age setting



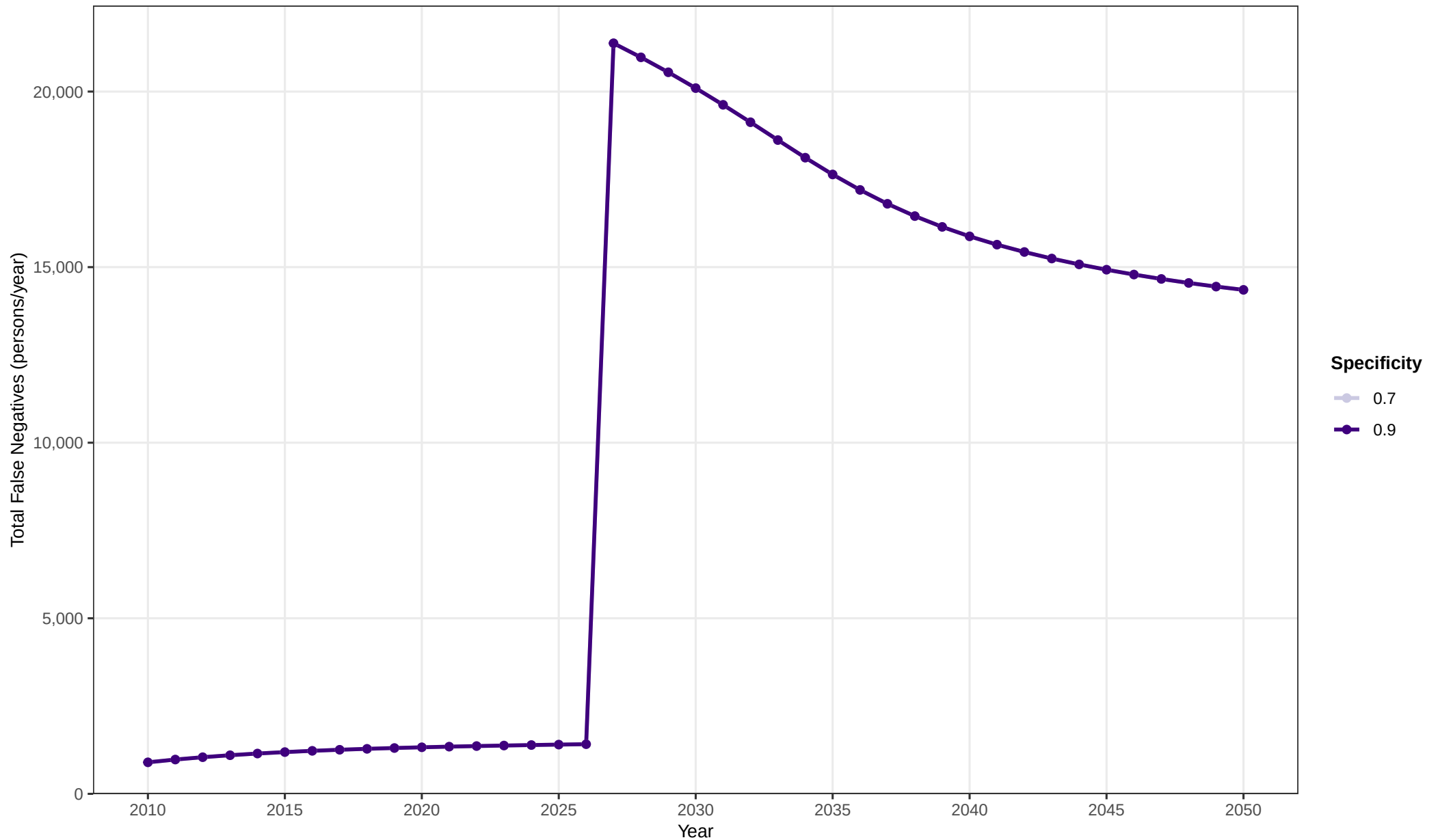
Total False Negatives (persons/year)

Mean by Screening Sensitivity setting



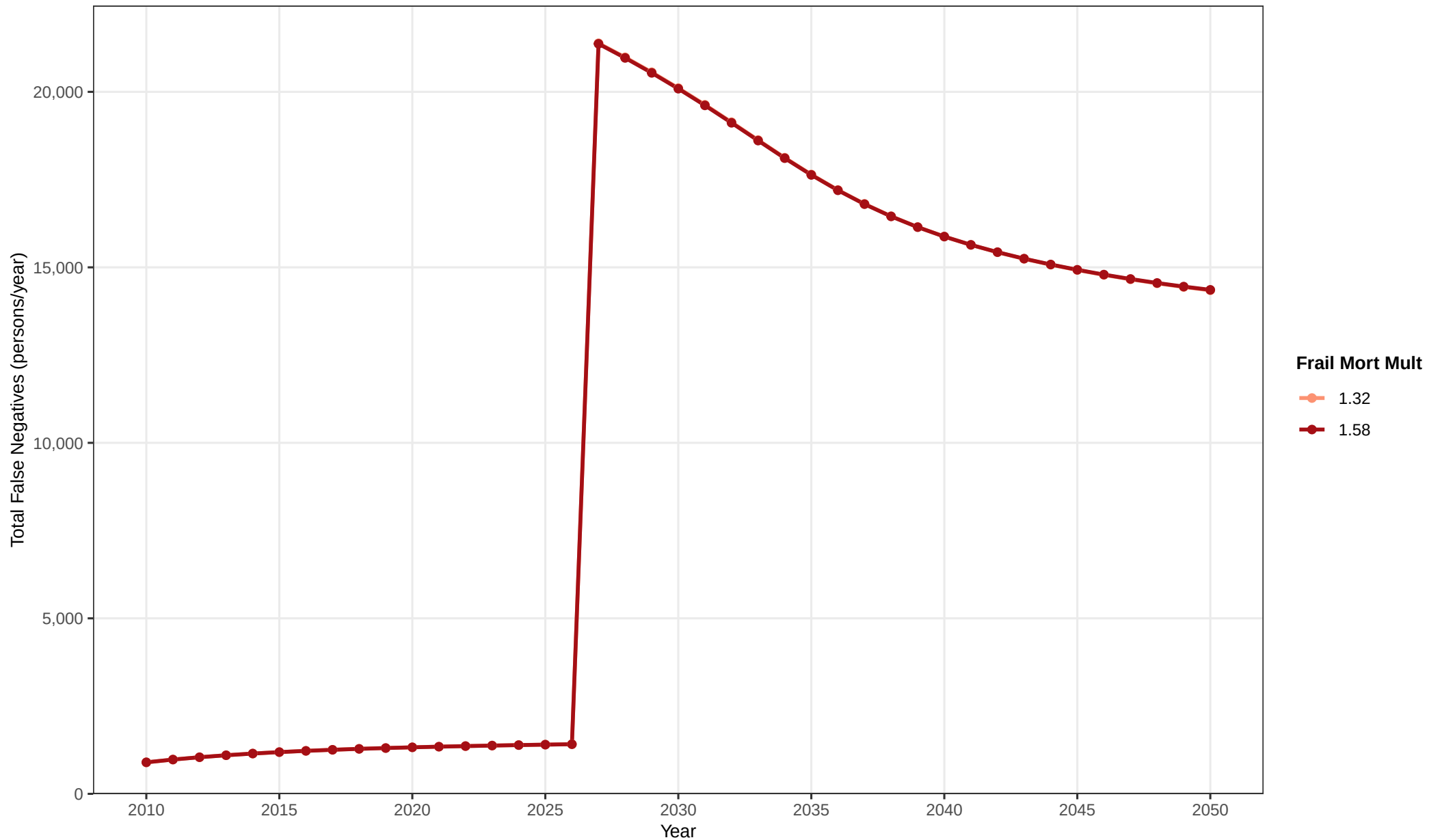
Total False Negatives (persons/year)

Mean by Screening Specificity setting



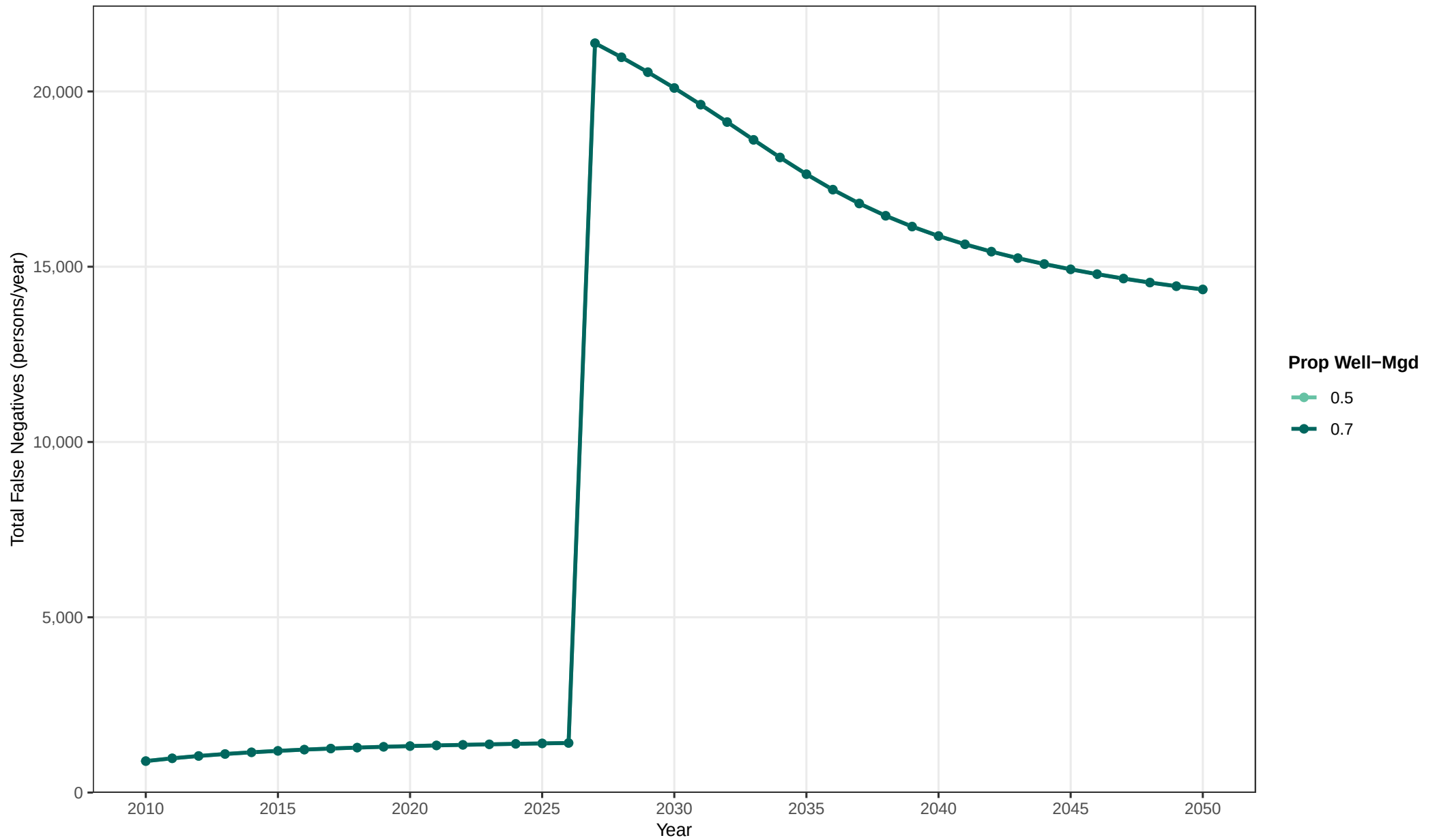
Total False Negatives (persons/year)

Mean by Frail mortality multiplier setting



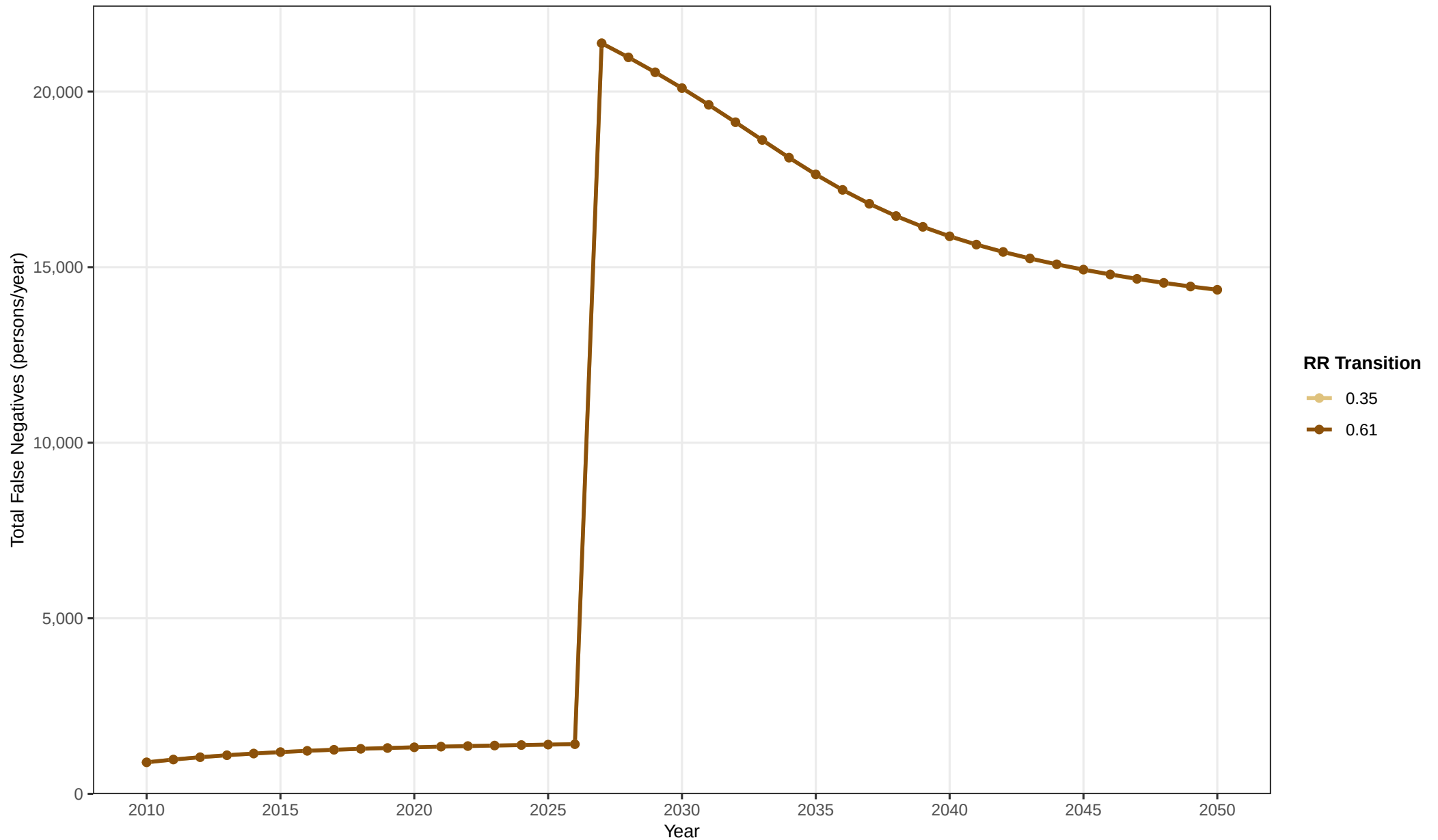
Total False Negatives (persons/year)

Mean by Proportion well-managed setting



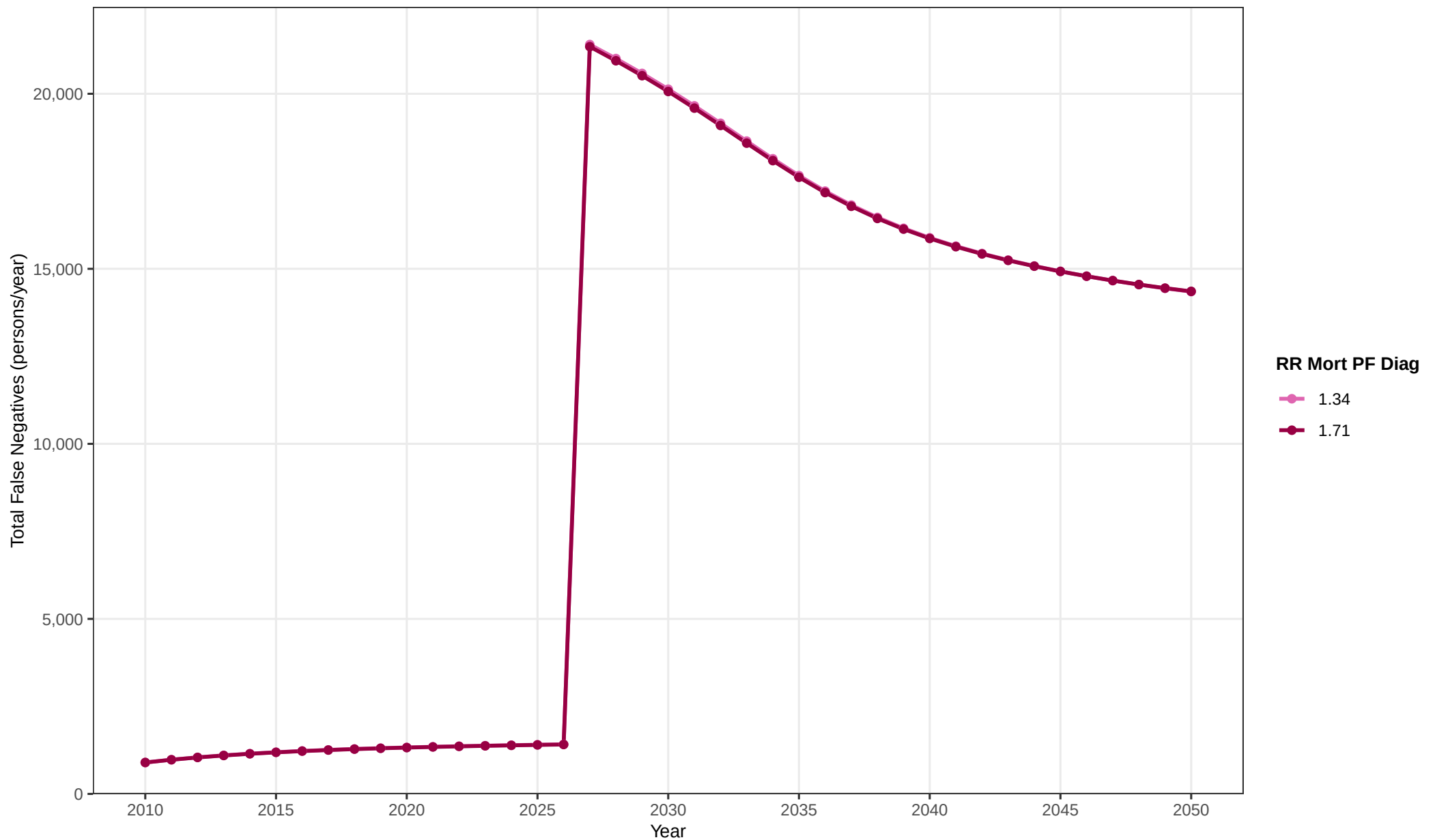
Total False Negatives (persons/year)

Mean by Relative risk of transition pre-frail to frail setting



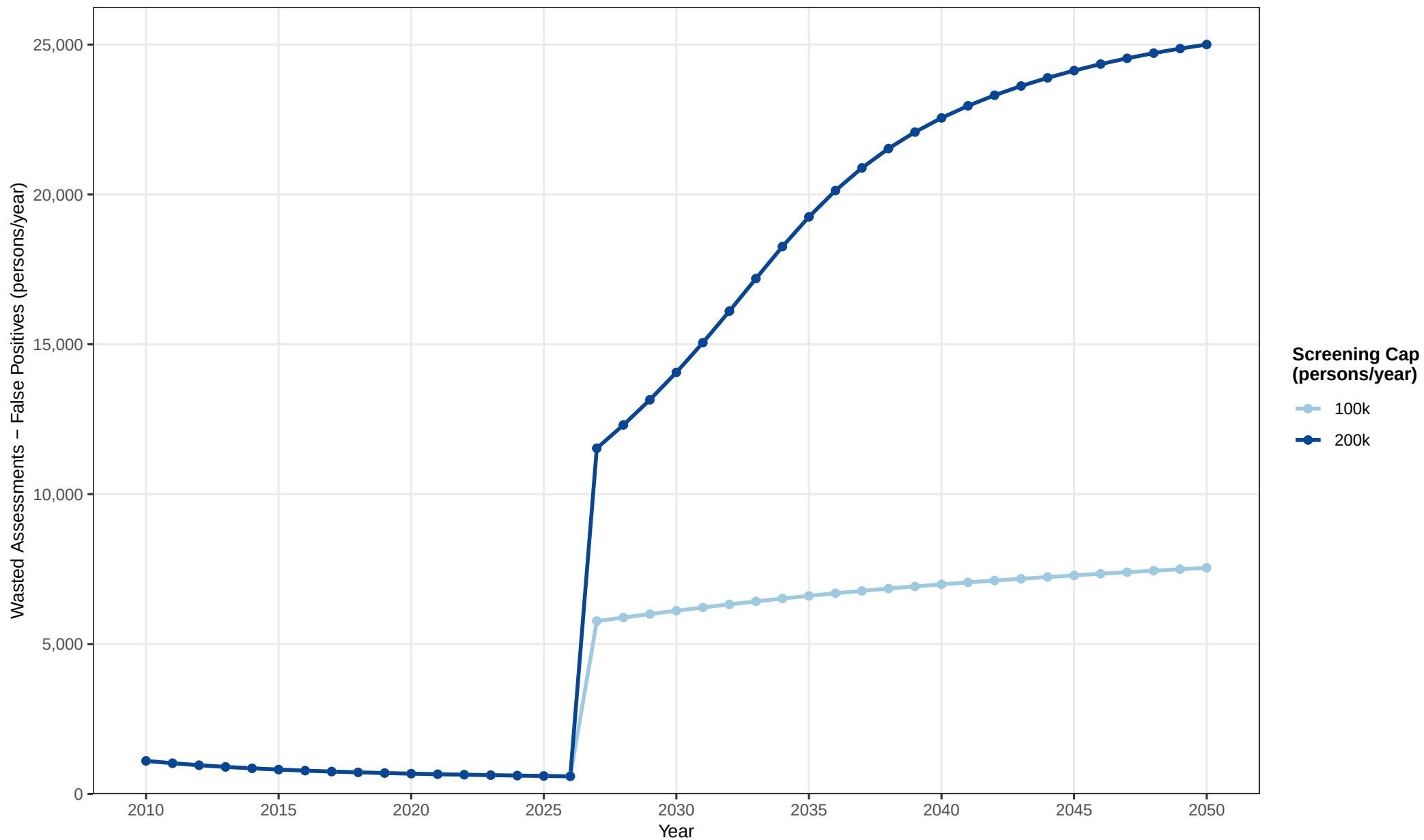
Total False Negatives (persons/year)

Mean by Relative Risk of mortality for pre-frail setting



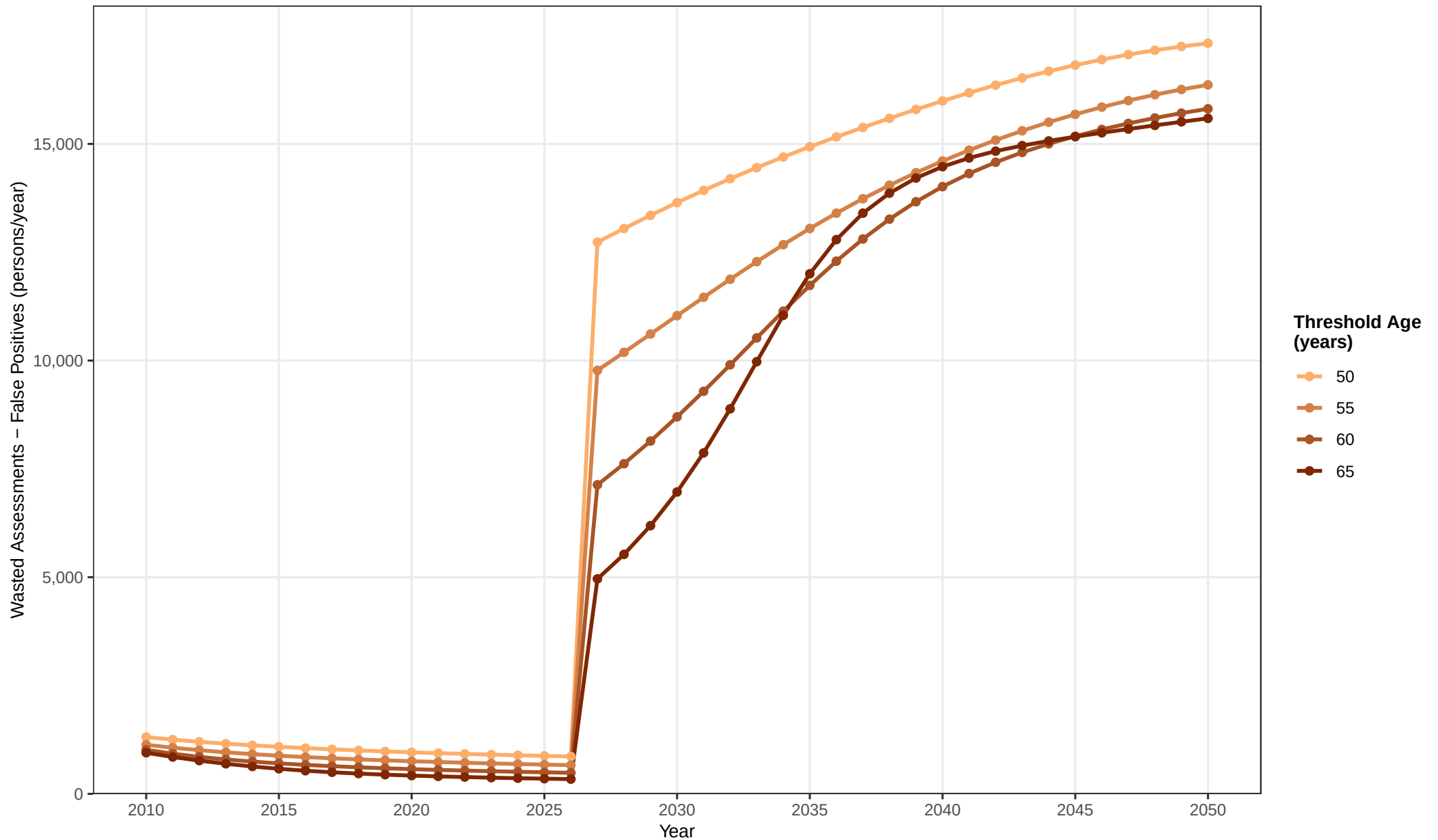
Wasted Assessments – False Positives (persons/year)

Mean by Screening Cap (Universal) setting



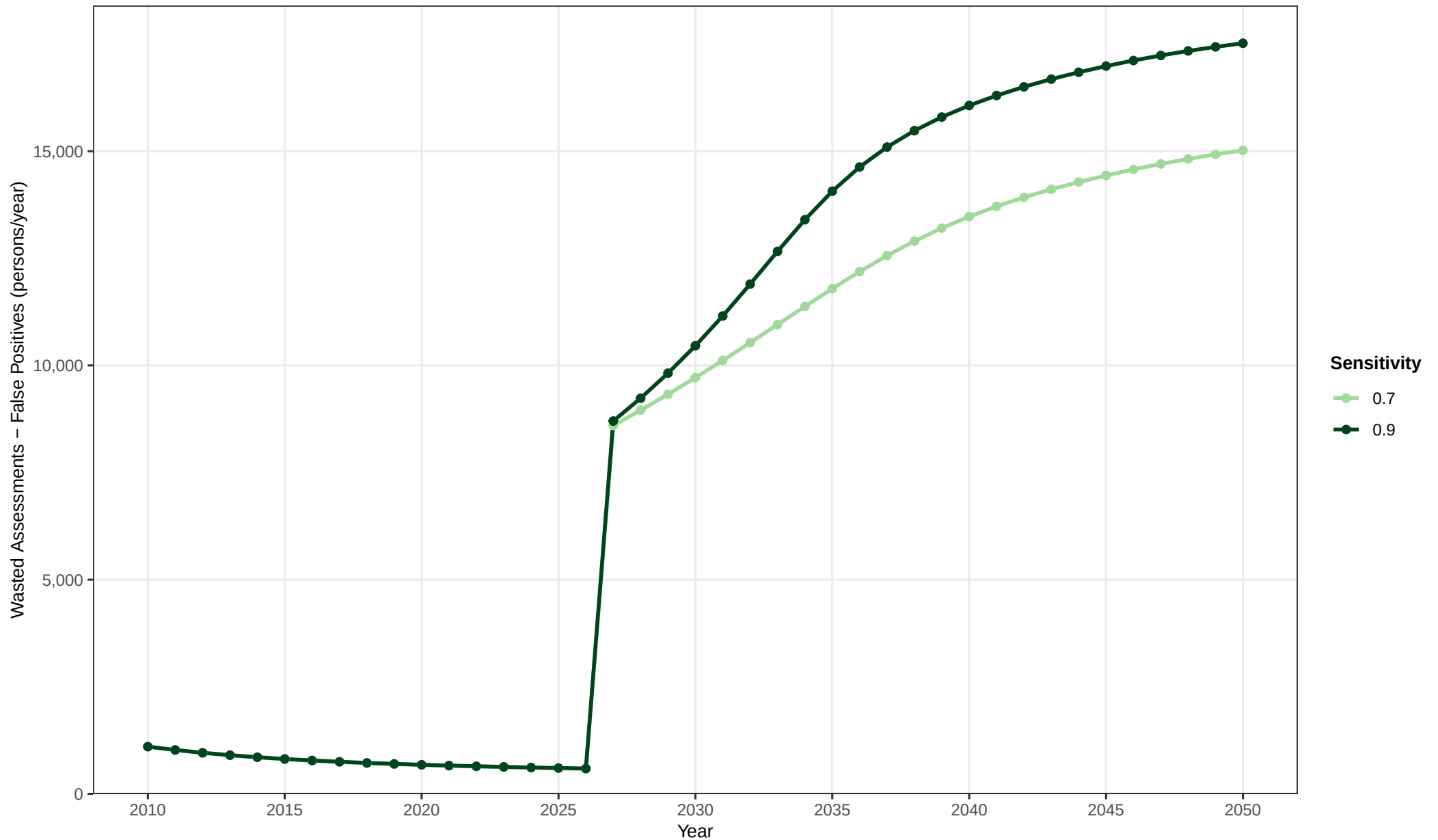
Wasted Assessments – False Positives (persons/year)

Mean by Screening Threshold Age setting



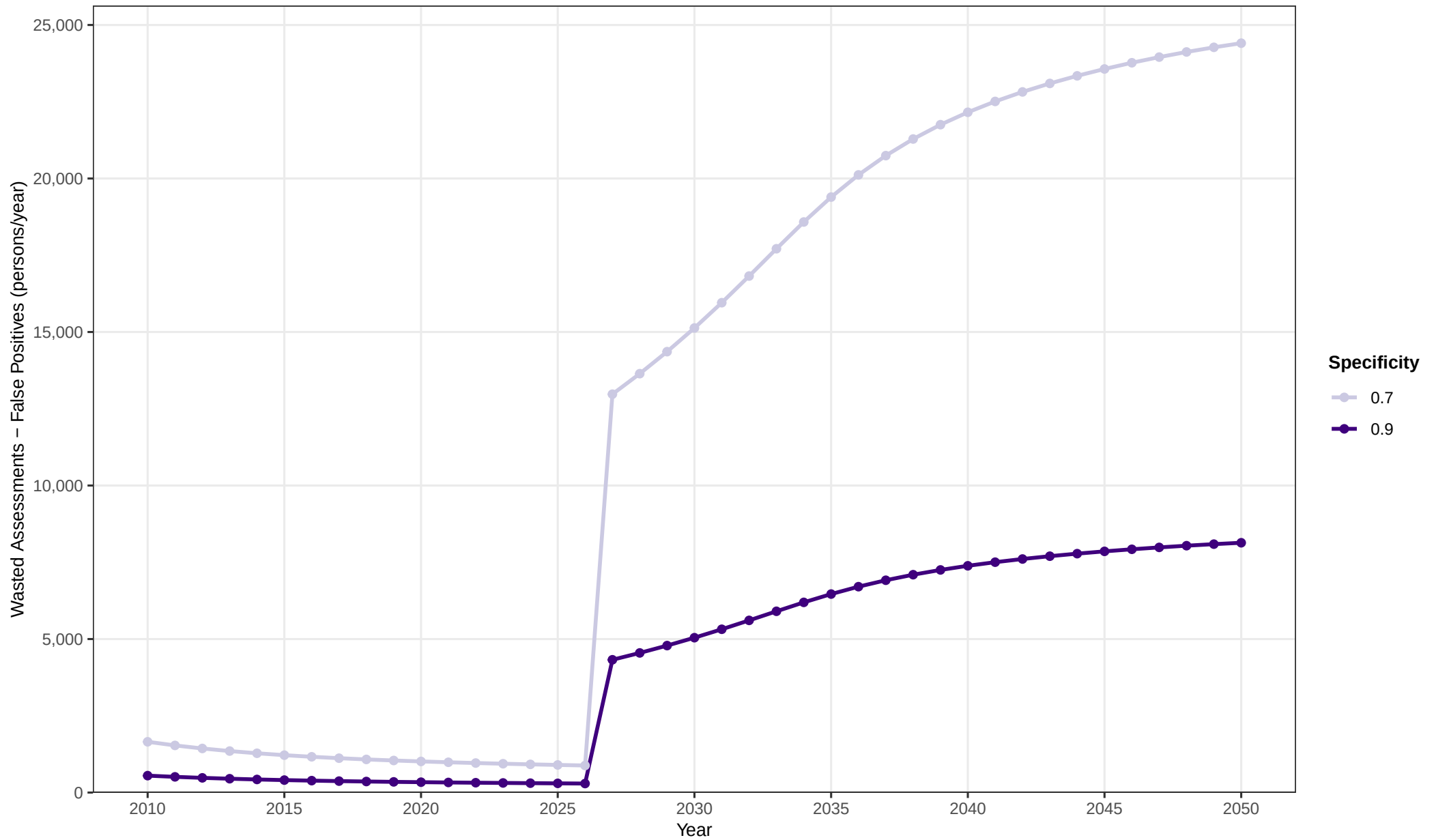
Wasted Assessments – False Positives (persons/year)

Mean by Screening Sensitivity setting



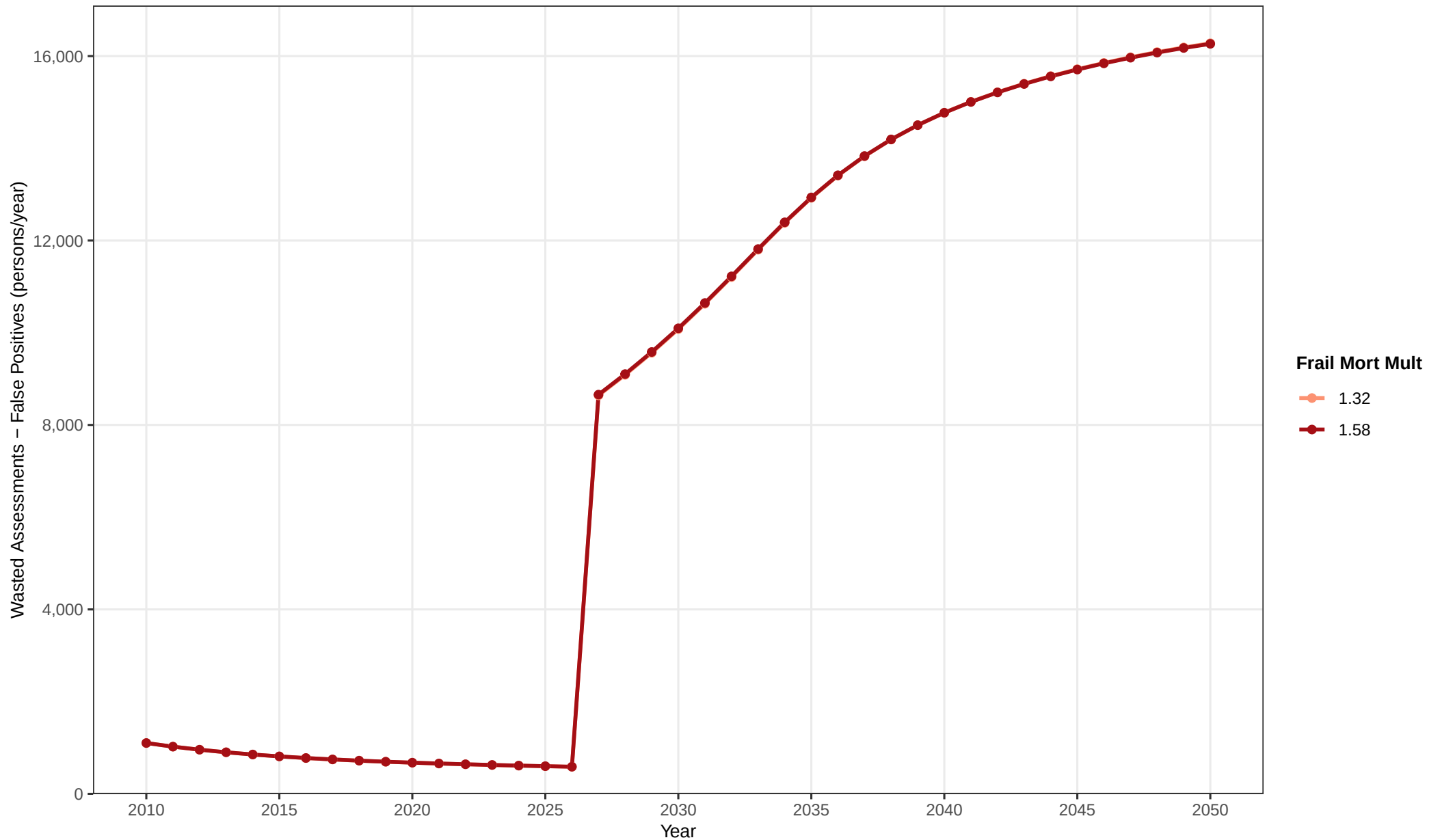
Wasted Assessments – False Positives (persons/year)

Mean by Screening Specificity setting



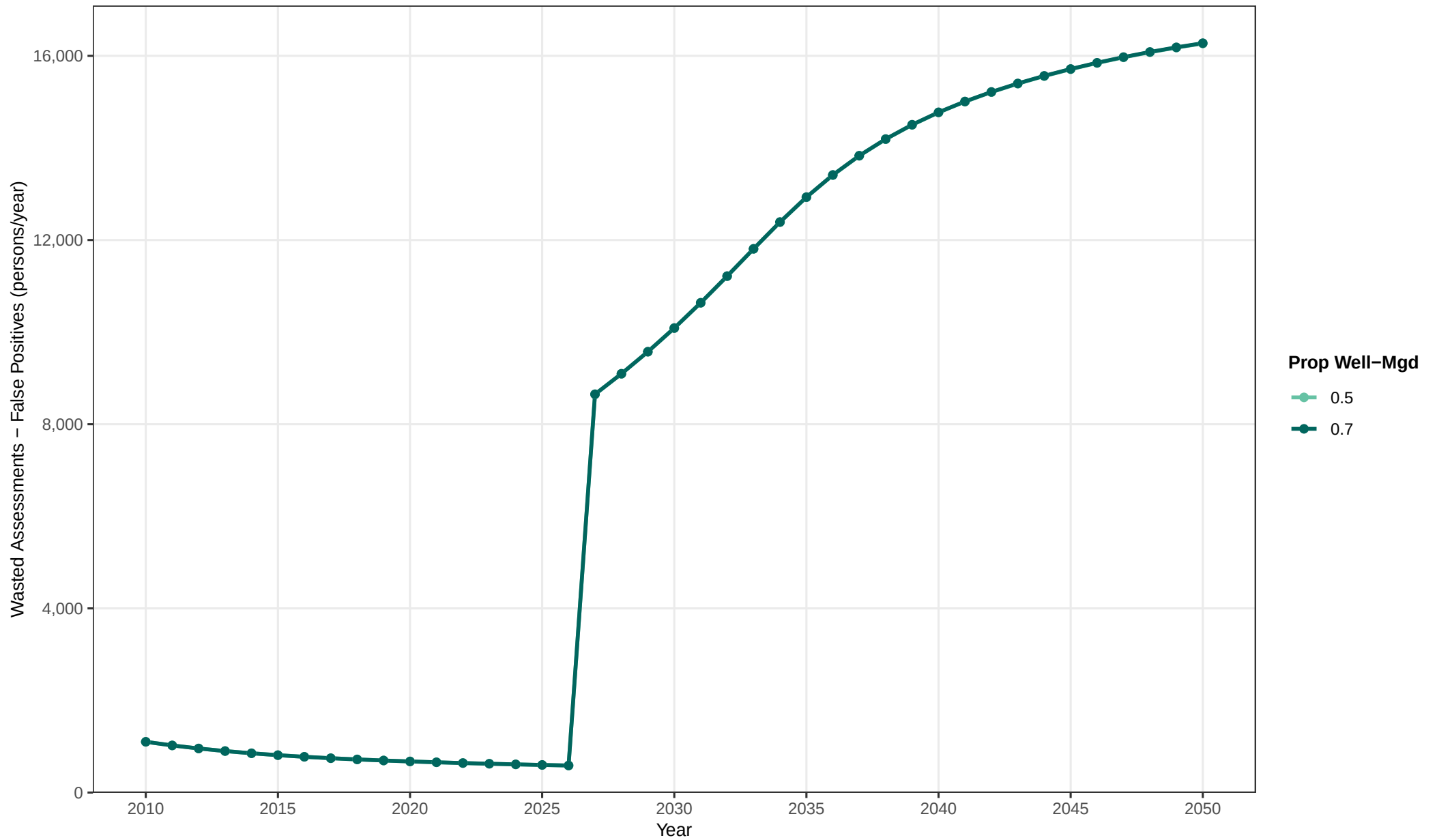
Wasted Assessments – False Positives (persons/year)

Mean by Frail mortality multiplier setting



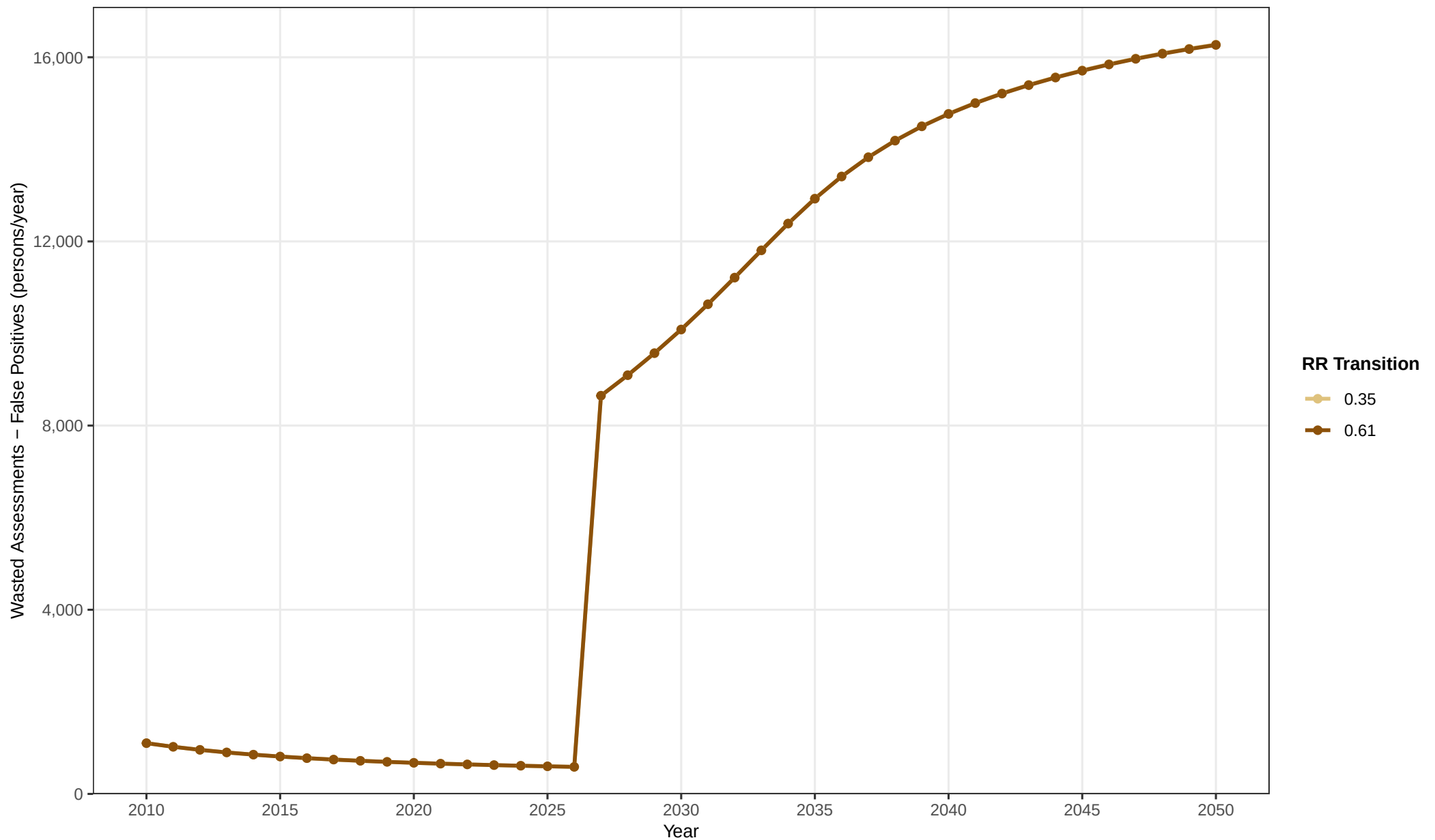
Wasted Assessments – False Positives (persons/year)

Mean by Proportion well-managed setting



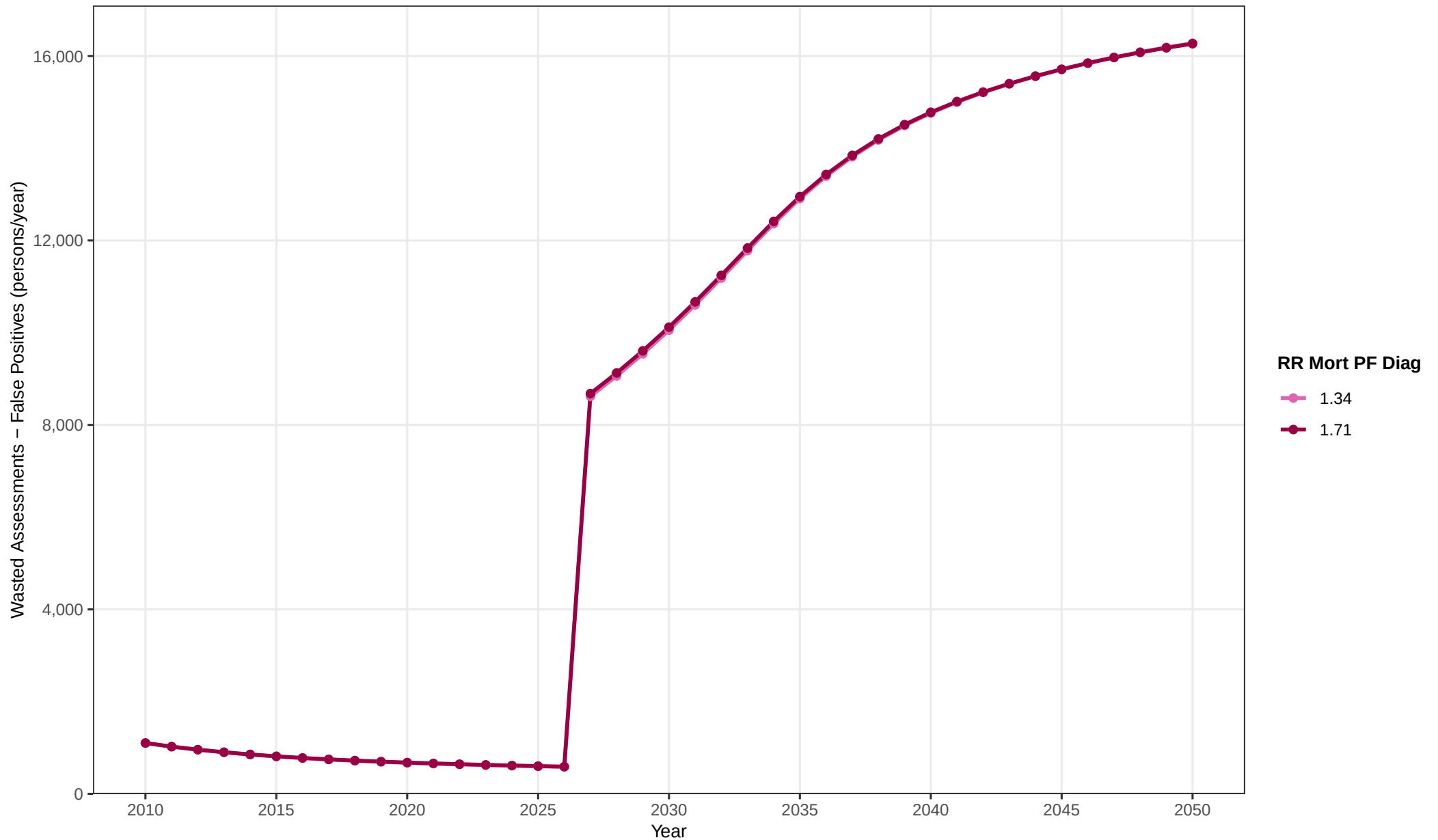
Wasted Assessments – False Positives (persons/year)

Mean by Relative risk of transition pre-frail to frail setting



Wasted Assessments – False Positives (persons/year)

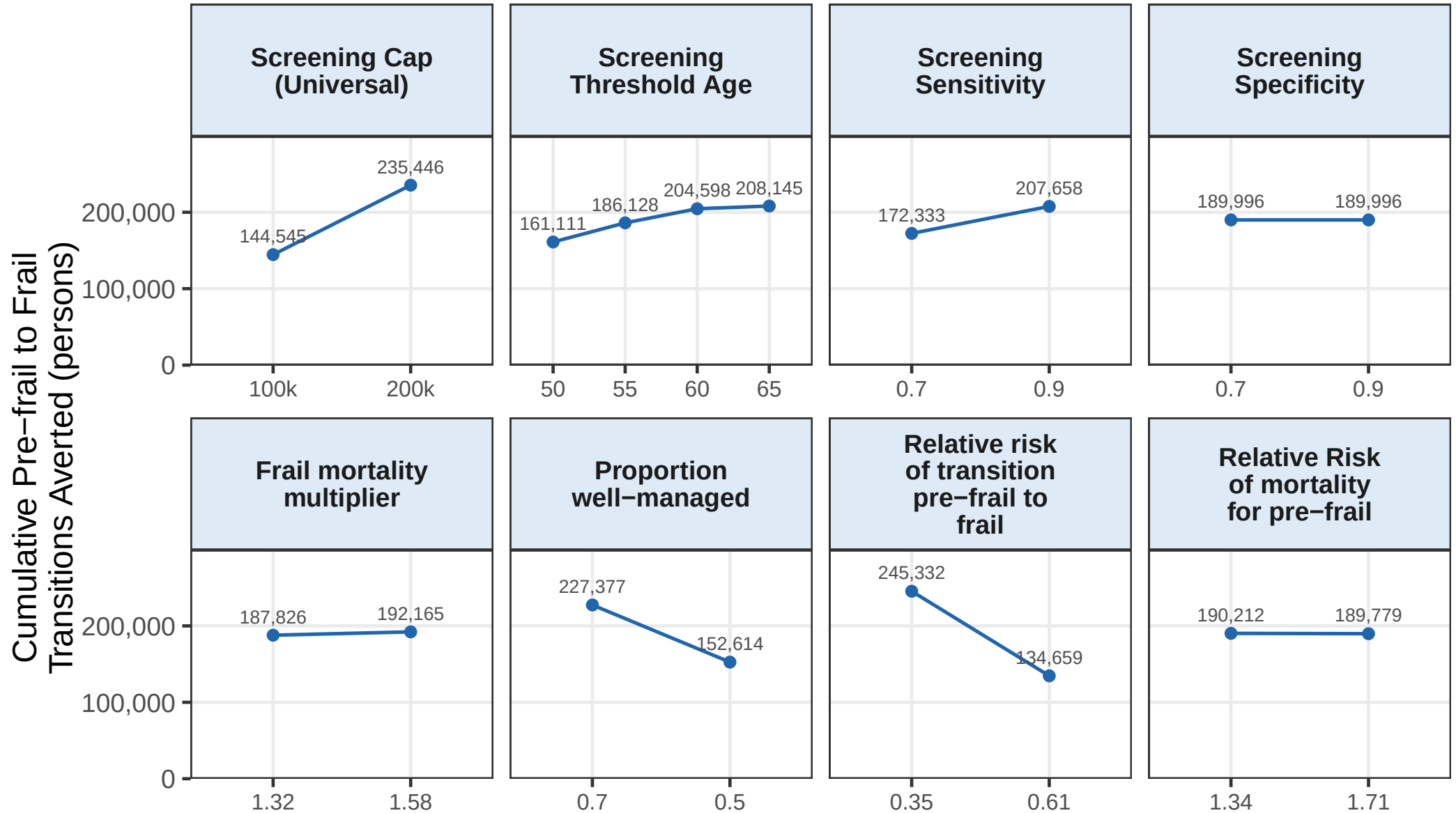
Mean by Relative Risk of mortality for pre-frail setting



III. Design of Experiments analysis

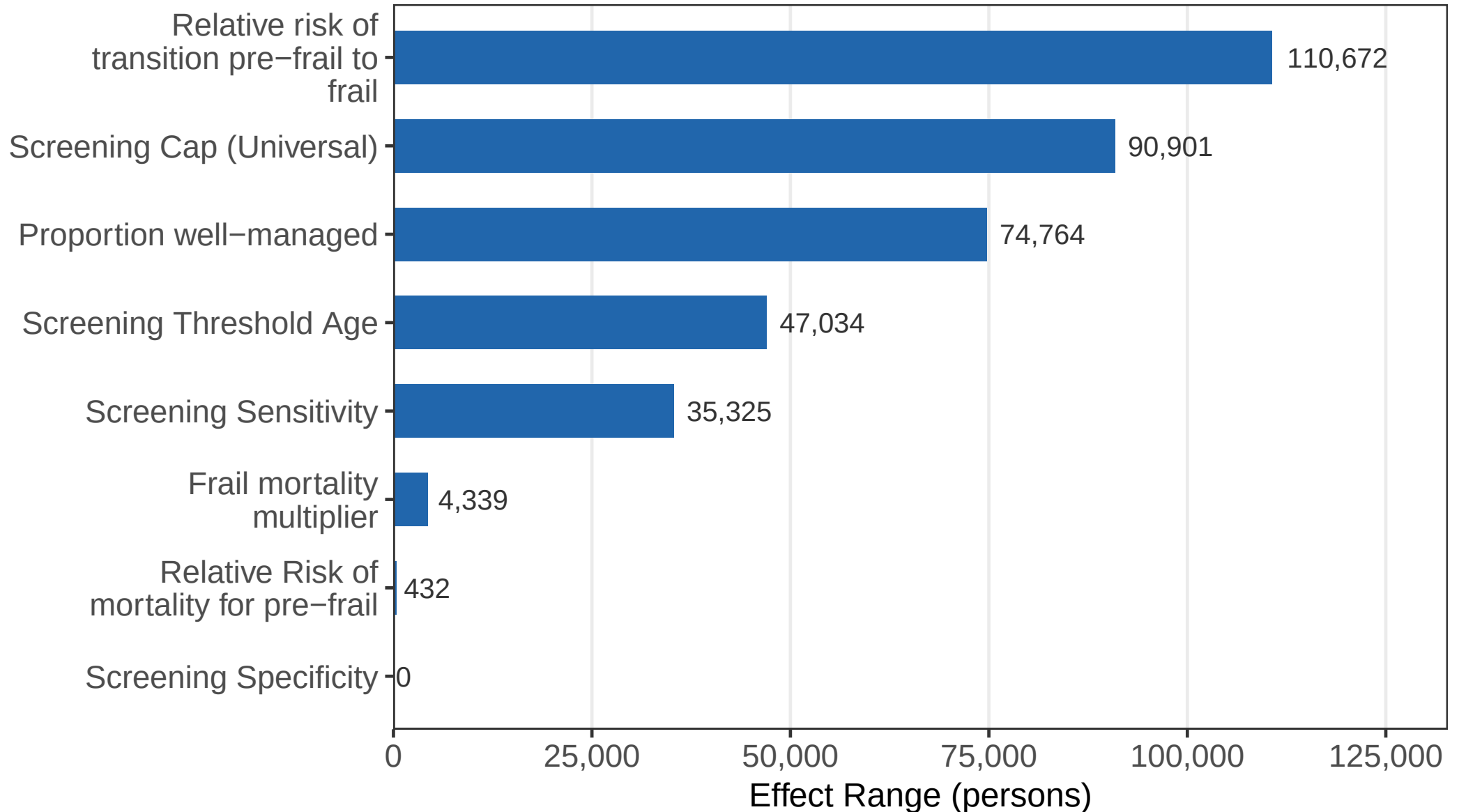
Main Effects

Mean Cumulative Pre-frail to Frail Transitions Averted (persons) | Reference year: 2050



Main Effects – Influence Ranking

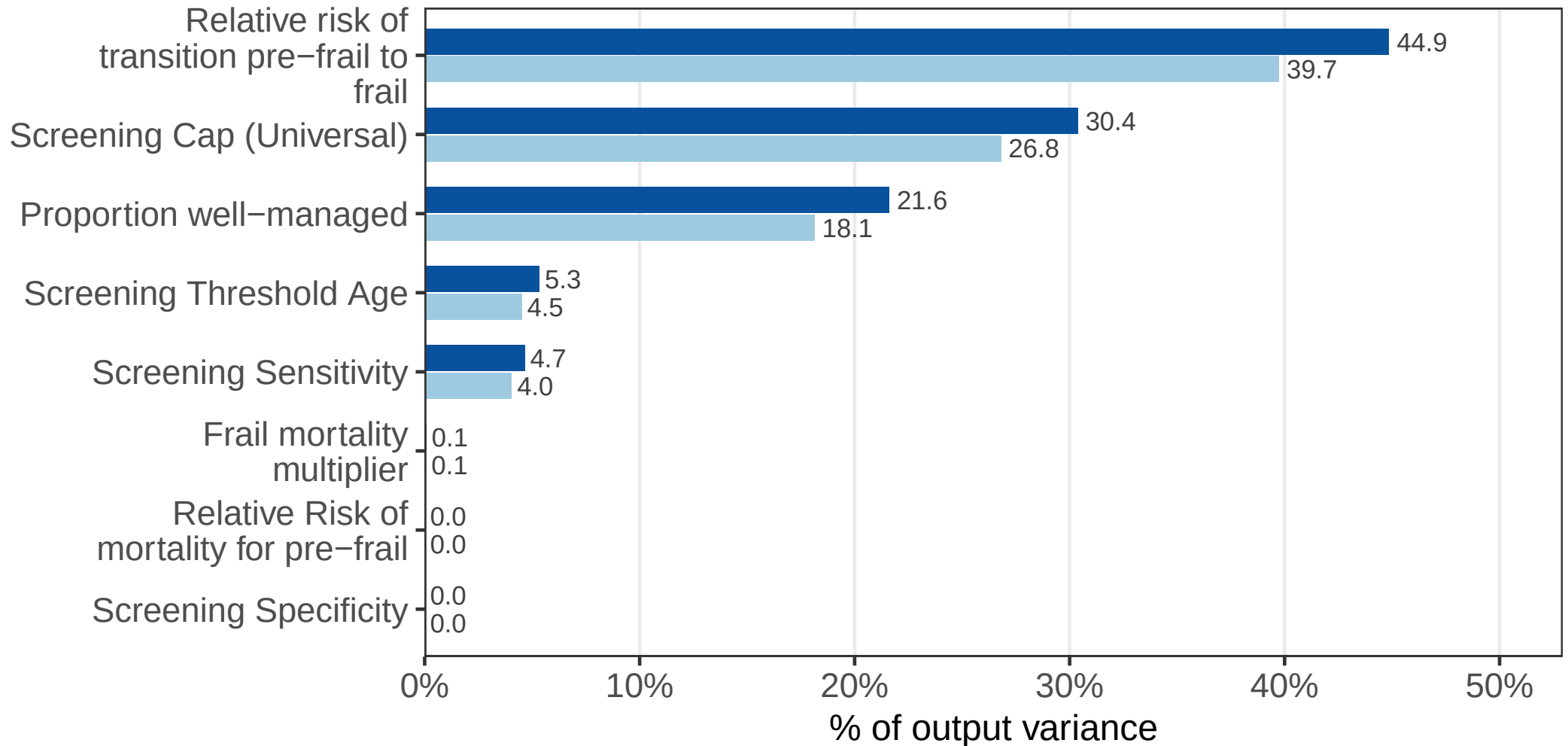
Effect range: max mean – min mean across factor levels



Global Sensitivity – Sobol Indices

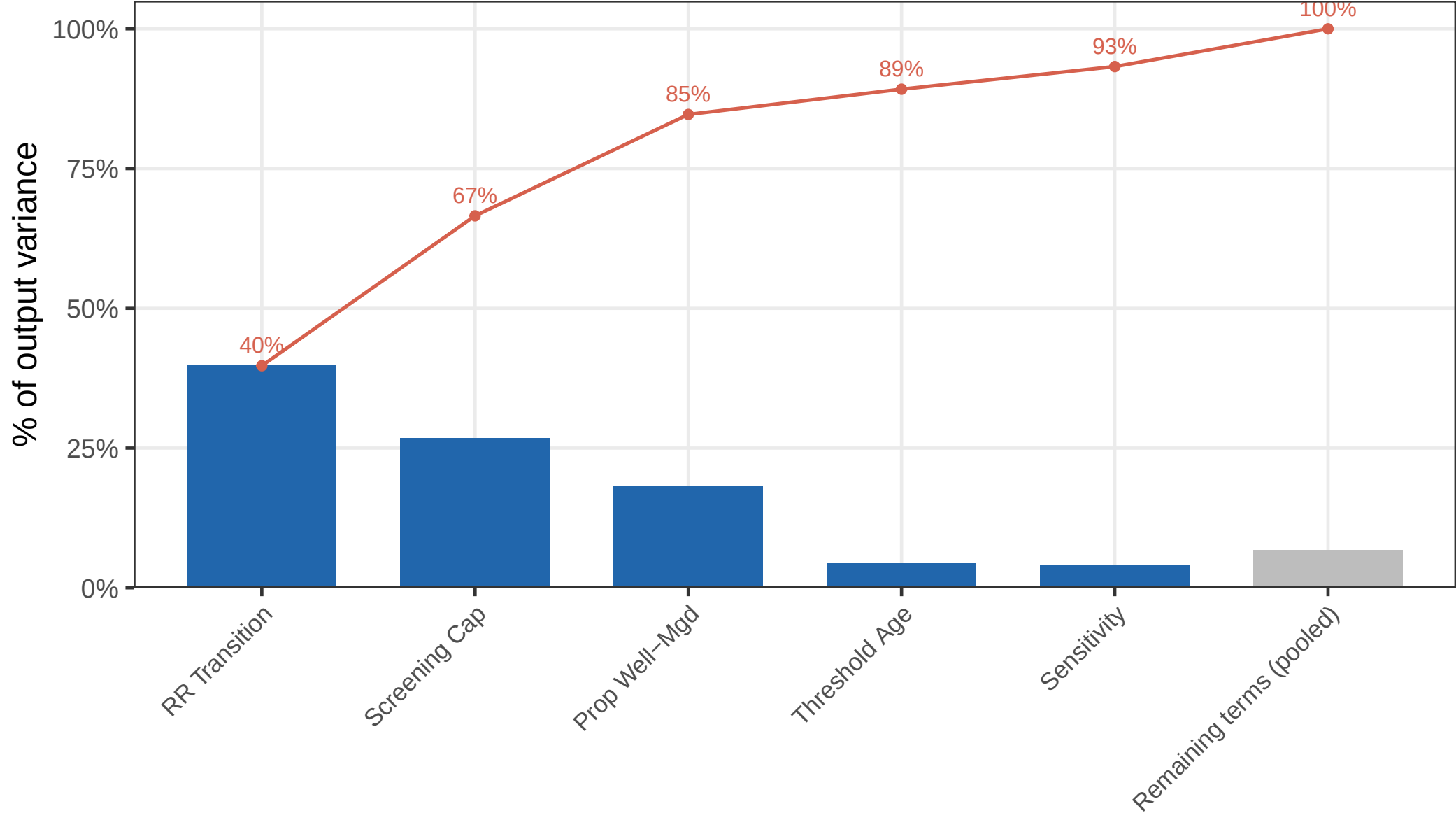
Share of variance in Cumulative Pre-frail to Frail Transitions Averted (persons) | Reference year: 2020
Gap between the bars = influence carried by interactions

First-order S_i Total-effect S_{Ti}



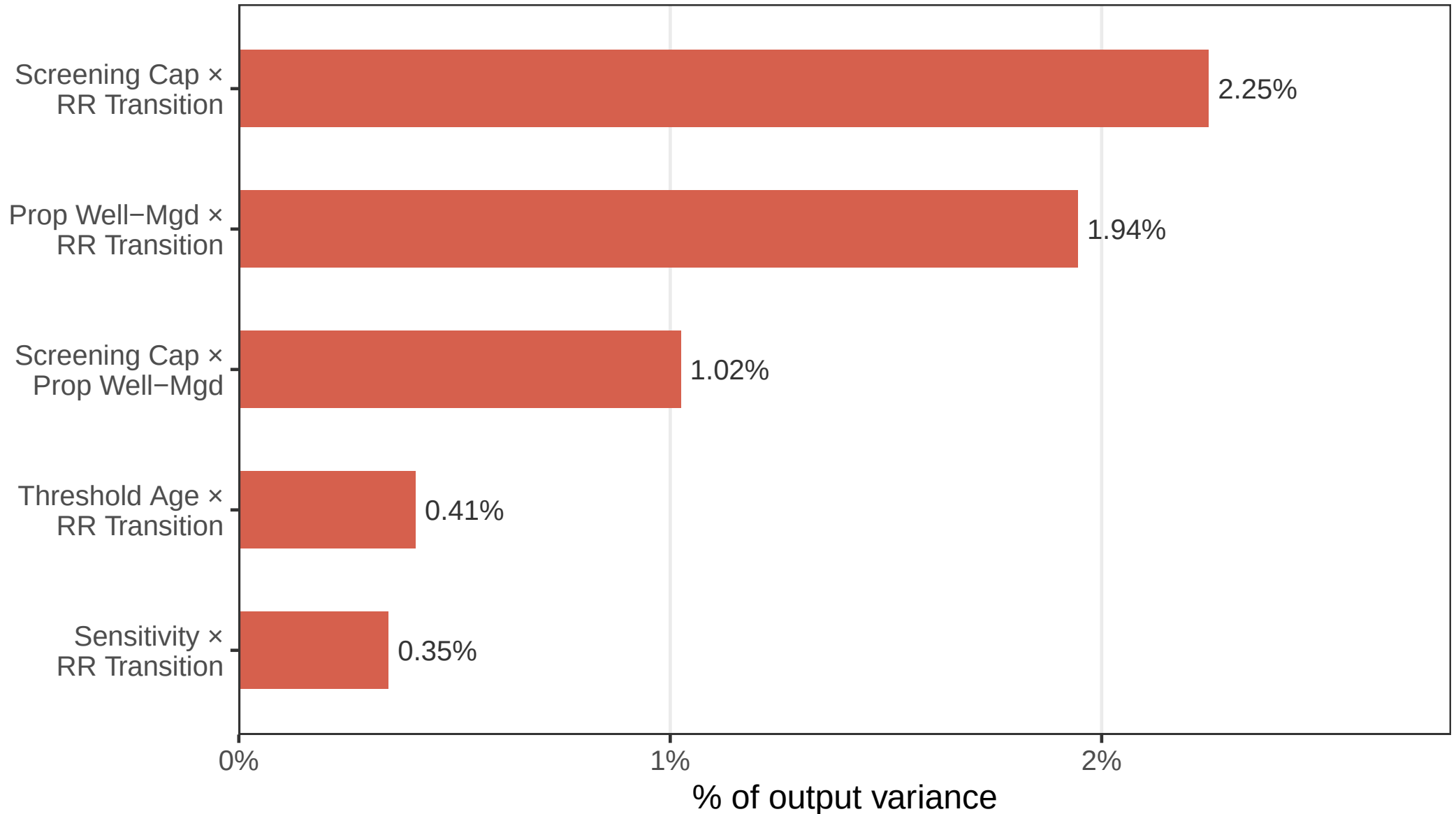
Variance Pareto – fitted terms by %SS

= term contribution; line = cumulative. Terms shown to 90% cumulative; rest pooled. Response: Cumulative Pre-frail to Frail Transitions At



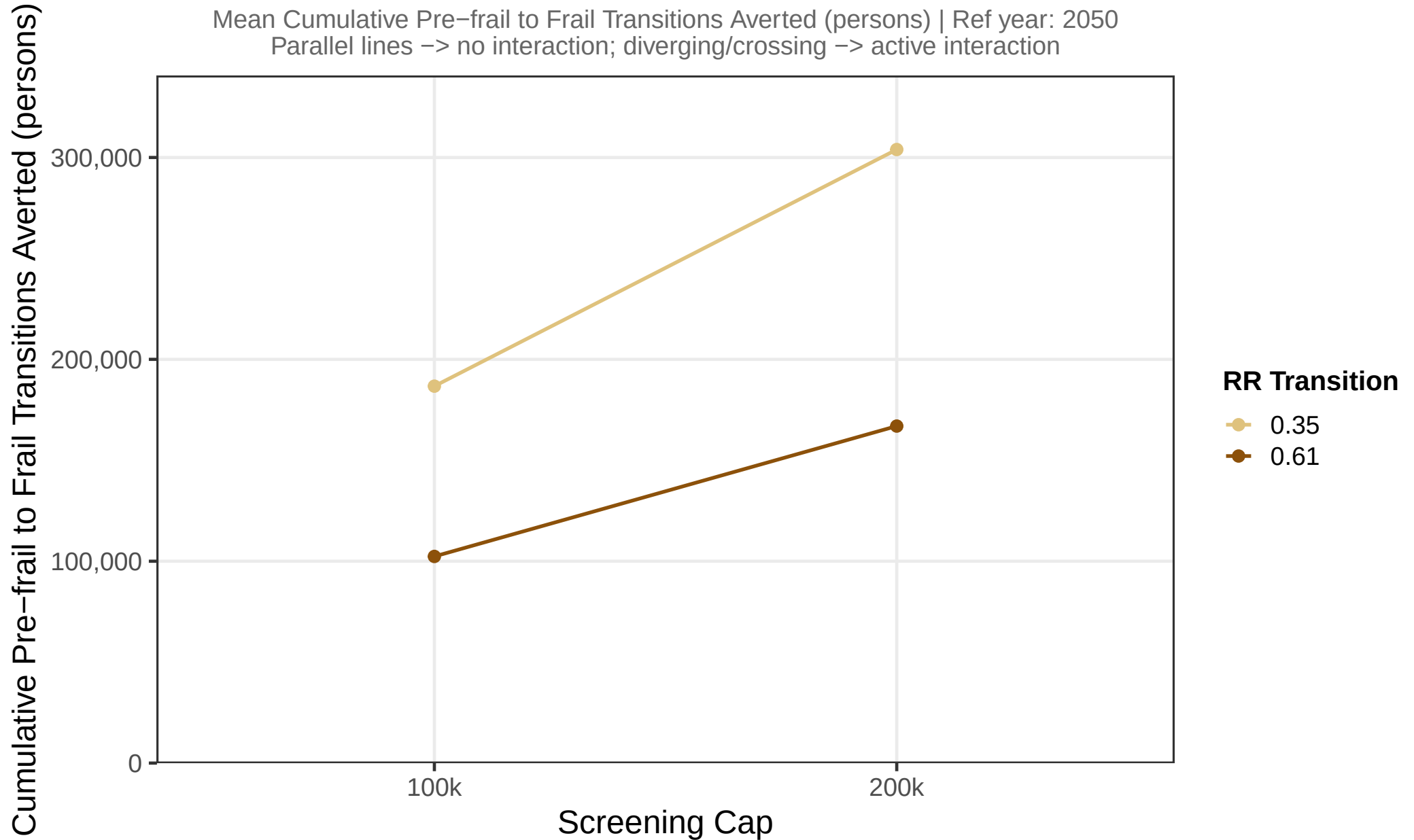
Interactions – Top 5 by Variance Share

2-way & 3-way terms ranked by %SS | Reference year: 2050



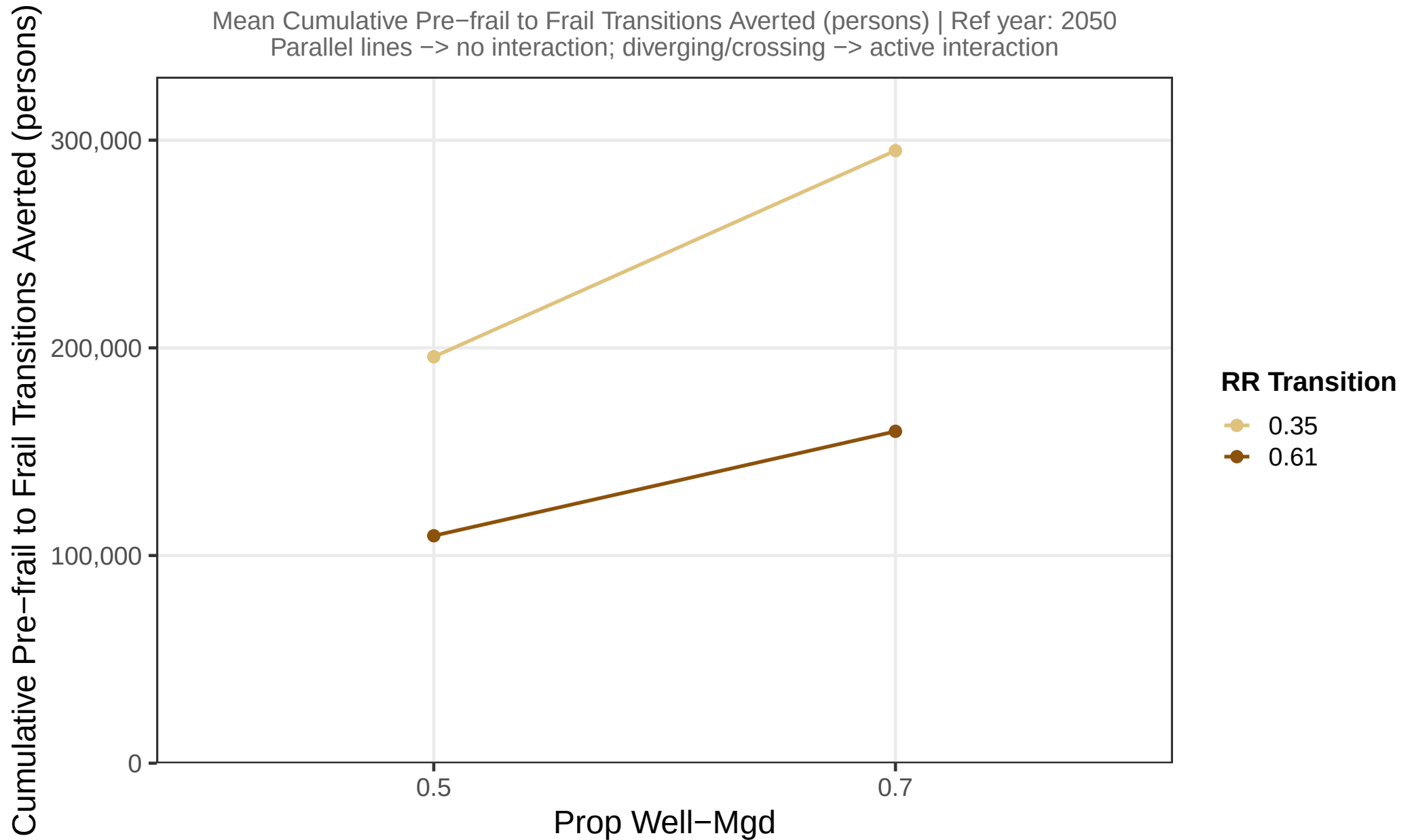
Interaction: Screening Cap × RR Transition

Mean Cumulative Pre-frail to Frail Transitions Averted (persons) | Ref year: 2050
Parallel lines → no interaction; diverging/crossing → active interaction



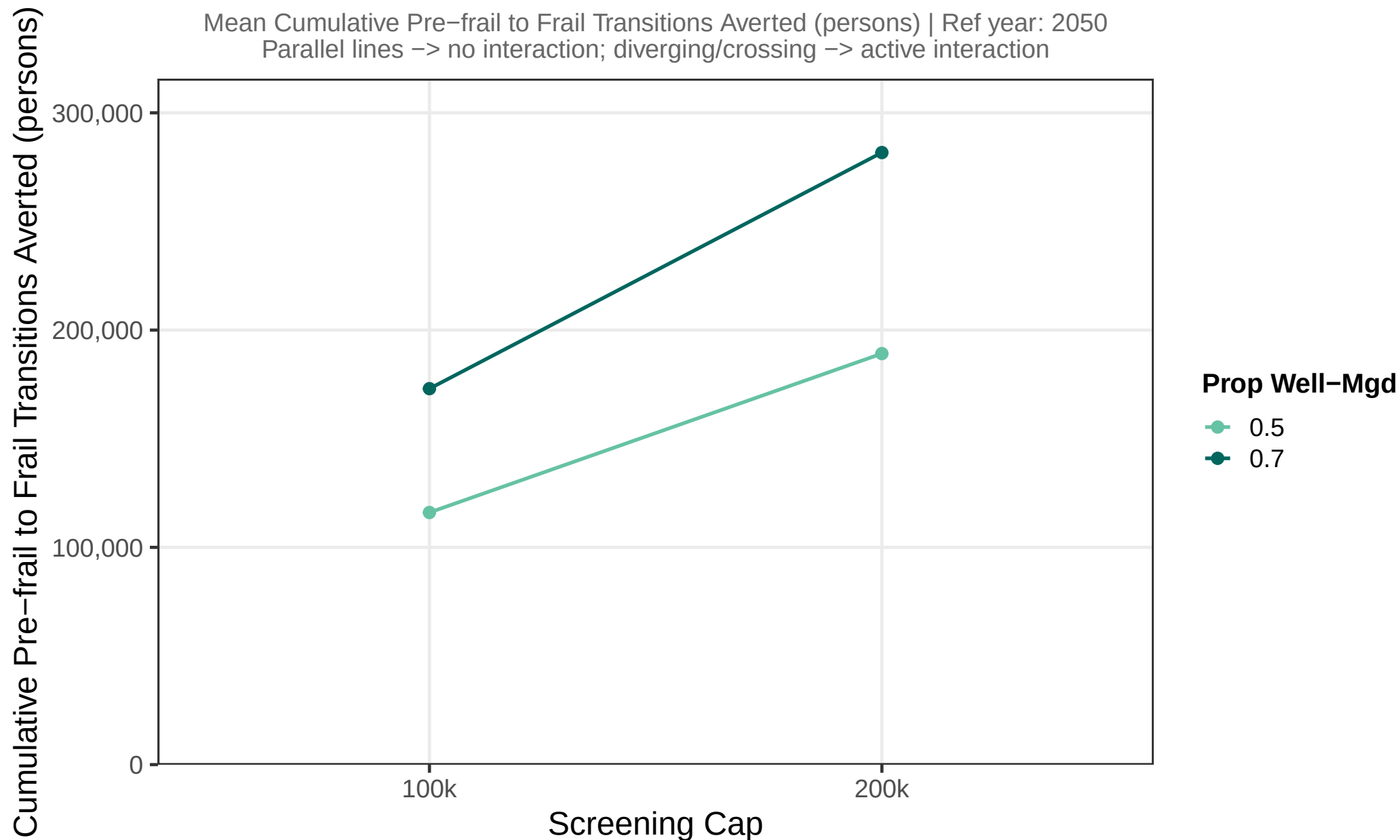
Interaction: Prop Well-Mgd × RR Transition

Mean Cumulative Pre-frail to Frail Transitions Averted (persons) | Ref year: 2050
Parallel lines → no interaction; diverging/crossing → active interaction



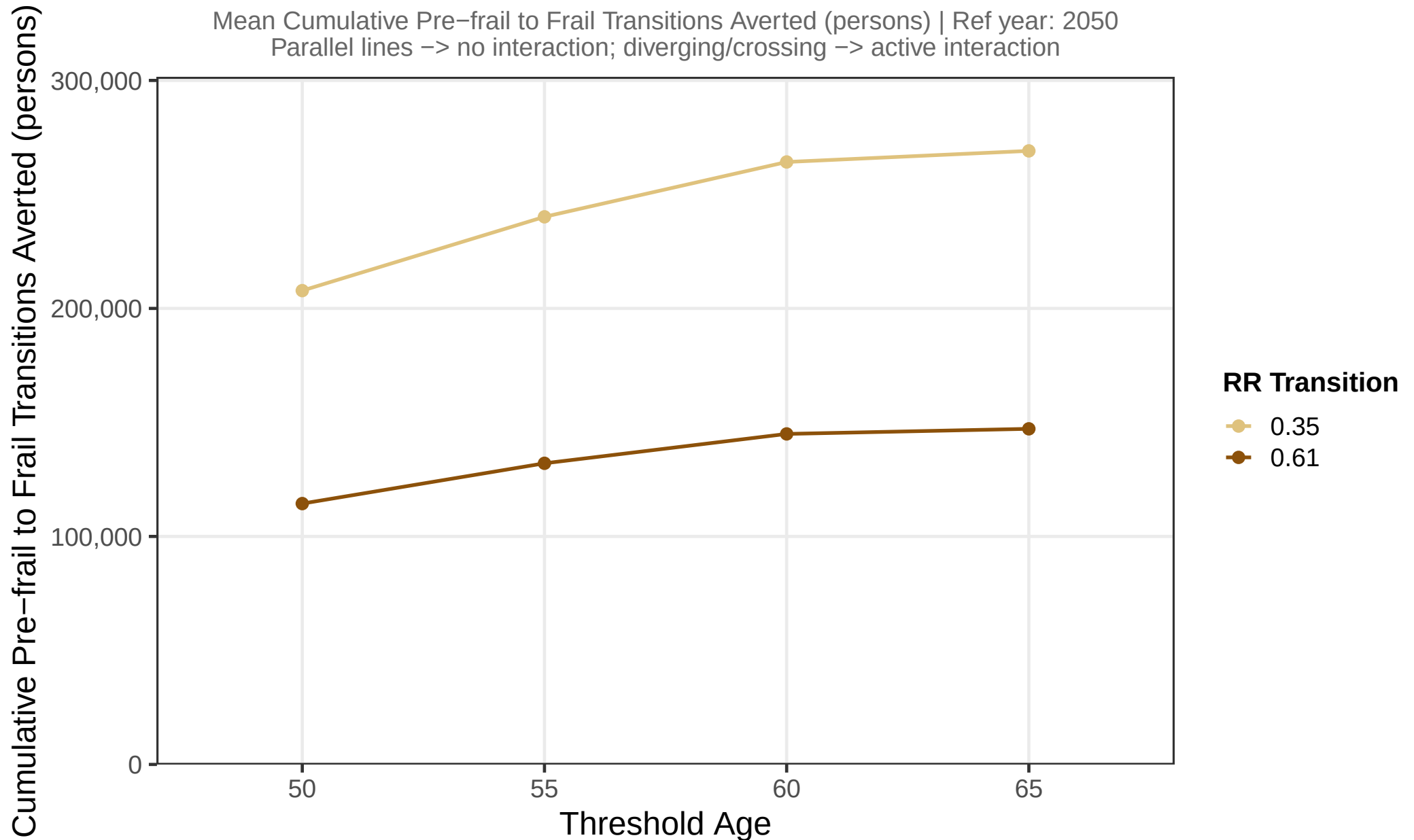
Interaction: Screening Cap × Prop Well-Mgd

Mean Cumulative Pre-frail to Frail Transitions Averted (persons) | Ref year: 2050
Parallel lines → no interaction; diverging/crossing → active interaction



Interaction: Threshold Age × RR Transition

Mean Cumulative Pre-frail to Frail Transitions Averted (persons) | Ref year: 2050
Parallel lines → no interaction; diverging/crossing → active interaction



Interaction: Sensitivity × RR Transition

Mean Cumulative Pre-frail to Frail Transitions Averted (persons) | Ref year: 2050
Parallel lines → no interaction; diverging/crossing → active interaction

